Worker Mobility and the Diffusion of Radical Technologies *

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Abstract

This paper studies how worker mobility and labor market institutions shape the diffusion of radical technologies through knowledge spillovers. I propose a theory with two-sided heterogeneity in which firms and workers accumulate technology-specific know-how through mutual learning, and worker mobility is subject to search frictions. When prior knowledge is only partially transferable to a newer technology, the reallocation of experienced workers becomes central to diffusing new technologies. I calibrate the model to U.S. labor markets in the context of a particular technology — predictive AI — to match empirical patterns using comprehensive microdata from LinkedIn. I then analyze the transition dynamics of introducing a radical new technology. Initially, scarcity of expertise forces firms to adopt internally using inexperienced workers. As diffusion progresses, the constraint on experienced labor relaxes, accelerating adoption through the reallocation of workers and generating an S-shaped diffusion curve. Aggregate output follows a J-curve pattern, with growth remaining sluggish initially before accelerating as productivity gains materialize. Poaching accounts for up to 60% of adoption, with its contribution peaking during the acceleration phase. A counterfactual exercise re-calibrated to European-style labor markets yields modestly higher steady-state output relative to the U.S., but adoption times two-thirds longer — highlighting a trade-off between labor market stability in the steady-state and technological dynamism during the transition.

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1 Introduction

Economic growth often unfolds in waves, each triggered by a radical technological breakthrough followed by continuous incremental improvements (Bresnahan & Trajtenberg, 1995; Mokyr, 2002; Schumpeter, 1939). A defining feature of many such technologies is that their adoption and implementation depend critically on specialized knowledge. When expertise in these technologies is scarce, it can become a bottleneck to their diffusion. What drives the spread of technologies that are increasingly dependent on scarce human capital? Understanding the underlying mechanisms is crucial to determine whether and how policy can facilitate a faster transition.

This paper stresses the central role of labor markets and specifically worker mobility in shaping the diffusion of radical technologies — that is, technologies that offer substantial productivity gains but require the accumulation of specialized knowledge for successful implementation and efficient use. Because such knowledge is jointly embodied in firms and workers' human capital, mobile workers serve as critical vectors for knowledge transfer, particularly during early adoption when implementation uncertainty is highest. Beyond technical expertise, they bring tacit knowledge about adapting existing business practices to new technologies and help firms avoid costly mistakes by sharing lessons from failed experiments at previous employers.

To examine this channel, this paper makes three main contributions. First, I develop a general framework of technology adoption and knowledge diffusion in which entrepreneurs and workers jointly accumulate technology-specific know-how, and knowledge is only partially transferable when upgrading to newer vintages. Second, I apply the framework to a particular technology: Artificial Intelligence (AI) adoption in U.S. labor markets. I match joint patterns of technology adoption and worker flows using comprehensive microdata covering the near-universe of LinkedIn to quantify the role of worker reallocation and poaching. Third, I show how labor market institutions can have sizeable effects on diffusion patterns and aggregate welfare in a counterfactual exercise re-calibrated to low mobility and high tenure jobs as seen in many EU labor markets.

The paper begins by documenting stylized facts about the correlation of worker flows with the adoption of radical new technologies. To empirically establish descriptive patterns, I focus on a particular example of such a technology: the adoption of predictive AI across firms in U.S. metropolitan areas. Using comprehensive resume data from the near-universe of LinkedIn, I classify firms as adopters and non-adopters based on AI-related keywords in workers' job titles and job descriptions, then examine whether newly adopting firms disproportionately hire workers from previous adopters. Two key patterns emerge from the data. First, employing the *excess probability* framework of Cestone et al. (2023), I find that the likelihood of hiring from an AI-adopter relative to a non-adopter jumps by 15% at the time of adoption — revealing the demand for specific external human capital when implementing a new technology. Second, hiring concentrates among experienced rather than junior workers initially, suggesting that worker mobility serves as a channel for transferring practical implementation knowledge and not exclusively technical skills. This motivates a general theory with two central components: reallocation of scarce labor with technology-specific human capital and learning through knowledge transfer.

I develop a theoretical framework to study how worker mobility and learning jointly drive the diffusion of new, path-breaking technologies. The economy is endowed with a set of technology vintages, each characterized by a general productivity level and a per-period fixed cost. There are two types of agents – workers and entrepreneurs – who meet in a frictional labor market. Workers are heterogeneous in both their level of human capital, which they accumulate specific to a technology vintage, and their employment status. Entrepreneurs differ in their productivity which is specific to the technology vintage they currently operate. Entrepreneurs are either matched with a single worker or operate solo and post vacancies to find a profitable match. In both states entrepreneurs produce output, but employing a worker offers two advantages: it enhances production through complementarities in the worker's human capital and the entrepreneur's productivity, and it enables mutual learning through the gradual transfer of knowledge.[1]

An entrepreneur-worker match jointly decides whether to upgrade to a new technology vintage. When upgrading, the knowledge is only partially transferable from the older to the newer vintage. Combined with the mutual learning process, this partial transferability creates a central role for experienced workers and their reallocation in driving the diffusion of a new technology and its economy-wide implementation. Experienced workers contribute through two channels. First, their accumulated technology-

[1] Note that this creates a trade-off between learning and the opportunity cost of foregone production. Production complementarities favor matching high productivity entrepreneurs with high human capital workers while learning gains favor matching differently knowledgeable agents.

specific experience directly enhances production. Second, they facilitate the transfer of knowledge, allowing entrepreneurs to more quickly acquire technology-specific knowledge through mutual learning.

As the main exercise, I apply this framework to study the full transition path of the economy when a new technology vintage is introduced. When calibrated to U.S. labor market data in the context of AI, the quantitative model delivers a characteristic S-shaped diffusion curve — taking approximately 3 years to reach 5% penetration and 12 years to reach 50%. The shape emerges endogenously from the interaction between random search and binding skill constraints.[2] Early in the transition, scarce access to workers experienced in the new vintage constrains adoption. As diffusion progresses, the likelihood that entrepreneurs match with workers experienced in the new technology increases, thereby progressively relaxing the human capital bottleneck and accelerating diffusion. Eventually, the process decelerates: as agents in the economy become more knowledgeable in the new technology and the labor market sorts high-human-capital workers into high-productivity firms, successful poaching becomes more difficult and costly. Aggregate output follows a J-curve pattern, with output lagging behind adoption as firms and workers incrementally build up their productivity and human capital in the new technology vintage, consistent with prior empirical and theoretical literature on general purpose technologies (Brynjolfsson et al., 2021; David, 1990; Helpman & Trajtenberg, 1994).

Central to these dynamics, access to workers with new-vintage experience is a binding constraint in the calibrated version of the model: while nearly all firms would adopt if they had access to a worker experienced in the new technology, only 50% of potential adoption-enabling matches are actually formed early in the transition. To examine which types of worker flows are driving the diffusion process, I progressively decompose adoption into distinct channels. Consistent with the data, internal adoption without external hiring constitutes the smallest fraction. Only approximately 5% of all adoption occurs in the initial phase without external hiring. The remaining 95% of adoption involves external hiring, with employment-to-employment (E-E) transitions playing a large role early on while slower-moving unemployment reallocation becomes more prevalent later in the transition.

The speed of the transition crucially depends on the degree to which diffusion occurs via poaching.

[2] The latter is an endogenous model outcome and finds support in firm-level survey evidence (European Commission, 2020; Rammer & Schubert, 2022). The model is for a general technology including ones in which access to experienced labor is not a binding constraint, in which case the diffusion curve may be concave instead of S-shaped – placing importance on skill bottlenecks.

Intuitively, absent E-E flows and in the presence of skill constraints on adoption, a greater portion of diffusion occurs via hiring out of unemployment and internally.[3] However, unemployment is a slower process as firms need to wait for firm-worker relationships to separate and associated with human capital depreciation while internal adoption does not benefit from knowledge spillovers. I discipline the strength of the poaching channel using the excess probabilities measured using the LinkedIn resume data, which provides a key moment on the correlation between technology adoption and poaching from previous adopters.

I discuss the effects of different labor market institutions on diffusion patterns and output dynamics. Specifically, I apply the framework to European economies characterized by lower mobility and longer tenure jobs. This exercise is motivated by Draghi (2024), who documents that the difference in U.S. and EU economic performance can be primarily attributed to the Information and Communication Technologies (ICT) sector and the use of modern technologies.[4] I examine whether the model can account for these patterns and perform a counterfactual exercise where I re-calibrate labor market parameters to match lower mobility rates and longer job durations as seen in many EU labor markets, revealing a trade-off between labor market stability in steady-state and technological dynamism during the transition. While aggregate output in the old steady-state is modestly higher relative to the U.S. calibration, the diffusion of the new technology is slowed down substantially. The time taken to reach 50% penetration of the new technology increases by two-thirds in the less dynamic labor market, and the welfare gain from the arrival of the new technology is lower by 8.5% of initial steady-state output.

In the final part of the paper, I derive the key inefficiencies in the model and explore whether there is scope for policy to intervene to speed up the transition.[5] To this end, I study a specific policy instrument: a time-restricted subsidy paid to adopters of the new technology. There exists a set of welfare-enhancing subsidies which are optimally phased out quickly and restricted to early adopters. In contrast, persistent subsidies may inefficiently speed up technology adoption and reduce welfare. To provide intuition for

[3]Unemployment in the model can be viewed as a stand-in for various slow-moving factors of labor reallocation. For example, workers may quit for personal circumstances or for amenity values. Transitions into unemployment from adopted firms therefore act generally as a slow-moving increase in the supply of workers experienced in the new technology. An exogenous increase in the pool of technology experienced workers through university would yield qualitatively similar dynamics. See Appendix D.2.

[4]In traditional sectors such as manufacturing EU economies are on par or even outperform their U.S. counterpart (Draghi, 2024).

[5]The previous paragraph already mentioned the largest source of inefficiency: frictional labor markets. This section takes the labor market institutions as given and considers inefficiencies conditional on those.

this result, I characterize the social planner solution and derive two primary inefficiencies. First, agents do not internalize the positive spillover effects of their adoption decision on others through subsequent worker mobility. This effect is strongest early in the transition when the pool of non-adopters is largest. Second, the search framework induces standard congestion effects in the labor market. These externalities are increasing in the aggregate amount of vacancy posting, i.e. in the middle phase of the transition, when entrepreneurs most strongly compete for diffusive workers. The proposed subsidy alleviates the first inefficiency but exacerbates the latter. A time-restricted subsidy balances these forces.

Literature. This paper adds to three strands of literature. First, I build on the knowledge diffusion literature (Benhabib et al., 2021; Buera & Oberfield, 2020; Lucas & Moll, 2014; Luttmer, 2007; Nelson & Phelps, 1966; Perla & Tonetti, 2014; Perla et al., 2021; Sampson, 2015). While these foundational papers study how productivity-enhancing knowledge propagates through the economy and shapes long-run growth patters, they are largely agnostic about the specific channel of diffusion. Recent work by Jingnan (2023) and Koike-Mori et al. (2024) examine knowledge diffusion through the reallocation of inventors around a Balanced Growth Path.[6] In contrast, I stress the importance of implementors in the adoption of a radical new technology – i.e. new technologies which require specific knowledge to be accumulated to effectively implement them. Moreover, I characterize the full transition dynamics over the lifecycle of a technology and show how labor market institutions fundamentally shape these diffusion patterns. This transition focus generates both the S-shaped adoption curves and J-curve productivity patterns, while revealing how labor market institutions mediate the trade-off between stability in steady-state and technological dynamism during the transition.

Second, the paper extends the literature on coworker learning (Adenbaum et al., 2024; Herkenhoff et al., 2024; Jarosch et al., 2021). While these papers demonstrate how within-team knowledge spillovers shape individual productivity, wage dynamics, and labor market inequality, they abstract from firm-level technology adoption decisions and do not differentiate between incremental and radical innovations — two features central to my analysis. Without the radical technology component and firm adoption decisions,

[6]A different strand of the literature links worker mobility and in particular job creation and job destruction to firm dynamics and the overall business dynamism of an economy (Davis & Haltiwanger, 2014; Decker et al., 2016; Hopenhayn & Rogerson, 1993).

my model reduces to a framework similar to Herkenhoff et al. (2024).

Third, the paper contributes to work on vintage human capital (Adão et al., 2024; Chari & Hopenhayn, 1991; Jovanovic & Nyarko, 1996; Violante, 2002), General Purpose Technologies (GPT) (Aghion et al., 2002; Helpman & Trajtenberg, 1994, 1996; Jovanovic & Rousseau, 2005), and technological revolutions (Atkeson & Kehoe, 2007; Caselli, 1999). While the vintage human capital literature largely focuses on generational changes in the supply of skilled labor in an Overlapping Generations framework, I emphasize worker mobility as an important channel for accessing skilled labor and model the interaction between worker and firm-side vintage knowledge.[7] My concept of radical technology draws parallels to General Purpose Technologies while its introduction can be interpreted as a technological revolution. The focus in the present paper is on GPTs which rely on specific knowledge.

Outline. Section 2 documents stylized facts that motivate the theory. Section 3 develops the theory. Its predictions are quantified in Section 4 and applied to the case of the growing EU-U.S. divide. Section 5 derives the social planner problem and studies a particular subsidy on adoption. Section 6 concludes.

2 Empirical Motivation: Worker Flows and AI adoption

Before turning to the theoretical framework, I briefly examine descriptive patterns for a particular example of a radical technology: Artificial Intelligence (AI).[8] These stylized facts – particularly the correlation between worker poaching from prior adopters and technology adoption – motivate the model, and provide a key moment to target in the calibration. While the empirical context provides a concrete example, the theoretical framework itself speaks to a general set of technologies.

Two empirical patterns guide the analysis. First, survey evidence across multiple countries confirms that skilled worker availability constrains AI adoption. Approximately 50% of firms in the European enterprise survey on the use of technologies cite skill shortages as a major obstacle to AI adoption, comparable to concerns about financial costs and investment risks (European Commission, 2020). The Mannheim In-

[7]Empirical evidence in support of skill obsolescence and vintage human capital is shown in Hombert and Matray (2024) and Kogan et al. (2021).

[8]The context will be predictive AI which was the predominant form of AI adopted between 2010-2024.

novation Survey (Rammer & Schubert, 2022) further finds that 40-60% of large knowledge-intensive firms use hiring to access third-party know-how. These findings underscore how access to skilled labor shapes technology adoption decisions — a central element in our theoretical framework.

Second, I document the correlation between worker flows and AI adoption patterns in U.S. firms in digitally-intensive sectors.[9] I collect motivating evidence for whether firm's hiring practices change around the time of adoption and, if so, for which type of workers and from which type of firms, adapting the concept of *excess probabilities* from Cestone et al. (2023). This exercise, detailed below, provides suggestive evidence for the model's mechanism and a key empirical moment for calibration.

2.1 Data

The primary dataset is Revelio Labs, a workforce intelligence company that provides harmonized resume data for a near-universe of LinkedIn profiles. The data contain rich information on workers' employment history and job characteristics at a monthly frequency. Variables include job location (state, city, MSA), seniority levels, detailed occupational and industry codes (SOC and NAICS), firm identifiers with a cross-walk to Compustat as well as detailed job titles and job descriptions. Access is granted via the Wharton Research Data Services website. Revelio Labs and similar resume data have been shown to be a good representation of the U.S. high-skill, professional labor market (Babina et al., 2024; Hampole et al., 2025; Schubert et al., 2025) and consistent with aggregate employment and occupational statistics from CPS and ACS (Tambe, 2025). The data are particularly well suited to study the professional digital intensive workforce that is the subject of study in this paper.

2.2 Identifying AI adopters

I classify firms as AI adopters and non-adopters for each year in the sample. This is done in two steps and follows a large body of literature that uses labor market statistics for measuring AI adoption (Abis & Veldkamp, 2023; Acemoglu et al., 2022; Babina et al., 2024; Hampole et al., 2025; Kalyani et al., 2025). The

[9] Those are the tech, professional services and high-growth manufacturing industries, i.e. sectors most exposed to AI. Those are also workers who are most likely to be complementary to AI technologies and not substituted by them, at least in the first wave of AI. Note that they may be both complementary to in-house use of AI technologies but also to integrating outsourced AI tasks into the business, especially for larger firms.

first step utilizes the rich raw text of the self-reported job titles and descriptions by Revelio Labs to categorize approximately one million workers in the data as using AI based on technology-related keywords. In a second step, I use the firm identifiers provided by Revelio Labs to map AI workers to individual private firms and Compustat firms. Firms are then classified as an adopter at time t if they employed an AI worker in year t or before.[10] The set of all adopters is denoted by $\mathcal{A}_t$. A firm k at t is referred to as a previous adopter or previously adopting firm if $k \in \mathcal{A}_{t-1}$. A firm j is referred to as a newly adopting firm if $j \in \mathcal{A}_t$ but $j \notin \mathcal{A}_{t-1}$.

In total, there are three ways in the data in which a firm can fill an AI position. They can promote an existing worker to an AI role and retrain the existing workforce, they can poach skilled workers from other firms and thereby access the knowledge of third parties, and they can hire an AI worker out of university. To avoid issues related to mechanical relationships, some specifications exclude from labor flows those workers used to identify the adoption year of firms.

Measuring propensity to hire from previous AI adopters. What is the propensity and direction of worker flows from AI utilizing firms to other firms in the economy? To provide descriptive facts I specify the following Poisson model for job-to-job movers. Let $E_{ickj\ell t}$ denote an indicator which takes the value one if a worker i belonging to a group c experiences a job-to-job move from firm k to j in location ℓ at time t. In the baseline, let $D_{ikt} = \mathbb{1}\{k \in \mathcal{A}_{t-1}\}$ denote the indicator that takes the value one if i's origin firm k belongs to the set of adopters at time $t-1$. The probability of a move for worker i to a firm j is specified as

$$E_{ickj\ell t} = \exp\left\{\alpha_{cj\ell t} + \gamma_{cj\ell t} D_{ikt}\right\} \varepsilon_{ikj\ell t}, \qquad i \in c \tag{1}$$

with $\mathbb{E}\left[\varepsilon_{ikj\ell t} \mid D_{ikt}, c, j, \ell, t\right] = 1$. The term $\alpha_{cj\ell t}$ is a firm-location-job-mover specific effect that captures the time-varying natural propensity of firm j to absorb workers in set c. This defines the baseline hiring probabilities that will be the reference for constructing the excess propensities.

I then call $\gamma_{cj\ell t}$ the *excess probability* of a worker moving from an AI utilizing firm to firm j.[11] It is

[10] One concern may be that AI hiring does not necessarily correspond to AI usage. Hampole et al. (2025) document that AI hiring in Revelio Labs correlates strongly with other, more common measures of technology use such as survey-reported AI utilization in the Census Business Trends and Outlook survey (BTOS) and patenting in AI using the USPTO AI patent database.

[11] This is with some abuse, as strictly $\exp \gamma_{cj\ell t}$ corresponds to the excess probability. However, $\gamma_{cj\ell t}$ maps directly to continuous-

defined relative to a baseline probability of absorbing a worker from a non-AI utilizing firm. The excess probability nets out factors that affect a firm's hiring uniformly. For example, if a firm is expanding and scales up all forms of hiring by a constant factor the excess probability is unaffected. Further, the focus on the conditional probability of a worker moving controls for time-varying changes in the size of pools of AI adopters. Formally, I estimate (1) as the sample equivalent of the log-odds ratio, that is,

$$\gamma_{cj\ell t} \equiv \log\left(\frac{\Pr\left[E_{ickj\ell t} \mid D_{ikt} = 1, i \in c\right]}{\Pr\left[E_{ickj\ell t} \mid D_{ikt} = 0, i \in c\right]}\right).$$

In a second step, I estimate[12]

$$\hat{\gamma}_{cj\ell t+h} = \beta_h \mathbb{1}\{Adoption_{jt}\} + \sum_{l=1}^{2} X_{jt-l}\delta_l + \alpha_\ell + \alpha_n \times \alpha_{t+h} + \nu_{cj\ell t+h}, \tag{2}$$

where $\mathbb{1}\{Adoption_{jt}\}$ takes the value one if firm j is classified as an AI adopter at time t, α_ℓ are MSA fixed effects, $\alpha_n \times \alpha_{t+h}$ are 3 digit NAICS - time fixed effects and X are pre-adoption controls that include log sales per workers deflated by NIPA prices, log professional employment, and professional share. The coefficients β_h then capture how relative hiring propensities change with time since AI adoption h.

Intuitively, this regression traces how hiring flows are associated with the distance to the adoption year. For example, for a firm that relies exclusively on internal training or hiring out of university we would expect β_h to not significantly change over horizon h; i.e. there not to be a change in poaching behavior. In contrast, for firms which use poaching to access the talent pool of AI utilizing firms we would expect β_h to increase around the time of adoption.

2.3 Hiring from previous adopters increases around time of adoption.

The baseline specification defines the set c as all professional workers with an employer-to-employer (E-E) move in metropolitan MSAs in the U.S. who move from some firm k to a publicly traded firm j. Figure 1 shows the coefficients β_h for the distance to adoption year $h \in [-4, 5]$. There is a discontinuous jump in

time transition rates which will be important in the quantitative model.

[12] In Appendix C.4, I discuss the variation exploited in the regression as seen through the lens of the quantitative model in the subsequent section.

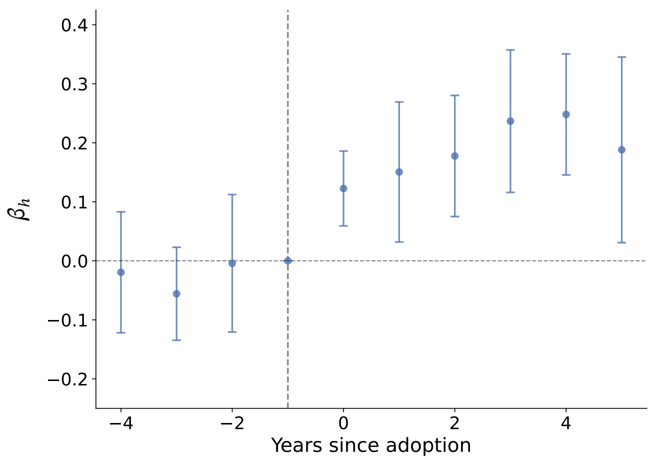


Figure 1: Excess probabilities.

Notes: This graph plots the coefficient estimates β_h from (2) and corresponds to how excess probabilities $\gamma_{cj\ell t+h}$ change with distance to the adoption year t. Estimated on subsample of E-E movers at annual frequency. The regression controls for industry-year and location fixed effects as well as pre-adoption variables related to selection into adoption (e.g. sales per worker). The sample period is 2010-2024. Destination firms in the sample are Compustat firms. Origin firms in the sample is the universe of firms in Revelio Labs. Standard errors are clustered at the firm-year level.

excess probabilities at the time of adoption of about 15% and a peak response of approximately 25%.

The discontinuity suggests that this correlation is specific to the timing of adoption events and does not capture smooth underlying trends. The increase in excess probabilities is persistent and approximately constant. Taken together, the descriptive evidence is consistent with the view that worker flows are important at the time of AI adoption and that AI adoption is accompanied by discrete changes in hiring practices to source of outside talent. Tapping into different talent pools seems to be a salient feature of the data even after controlling for productivity and factors at the industry-time level.

A baseline estimate of 0.15 implies a 15% increase in the log-odds ratio, translating to approximately $e^{0.15} - 1 \approx 0.162$ or a 16.2% increase in the relative probability that firms hire from previous adopter compared to their baseline hiring patterns. The magnitude of the baseline estimates compare to an interquartile range of excess probabilities in the data of about 0.8. The estimates are therefore meaningful but not unreasonably large. Importantly, these estimates capture changes across the entire professional workforce, not just inventors as previously studied in the literature. These estimates are robust to a series of variations explored in Appendix C.5.

One concern is that our adoption measure, employing AI workers, and our outcome, hiring from AI utilizing firms, both involve labor market measures, potentially creating a mechanical relationship. I address this in several ways. First, I re-estimate the above specifications excluding the first AI workers in a firm, i.e. the workers used to classify adoption events (see Appendix C.5.1). Specifically, Figure C.4 shows that estimates are both qualitatively and quantitatively robust to this further sample restriction. Second, insofar a mechanical relationship exists it would be concentrated at horizon $h = 0$. Effects are, however, strongly persistent and approximately constant. Third, our adoption measure is not determined by *cross-firm* labor flows necessarily. In total there are three distinct channels: (1) promotion of existing employees to AI roles (2) hiring from universities and (3) poaching from other firms.

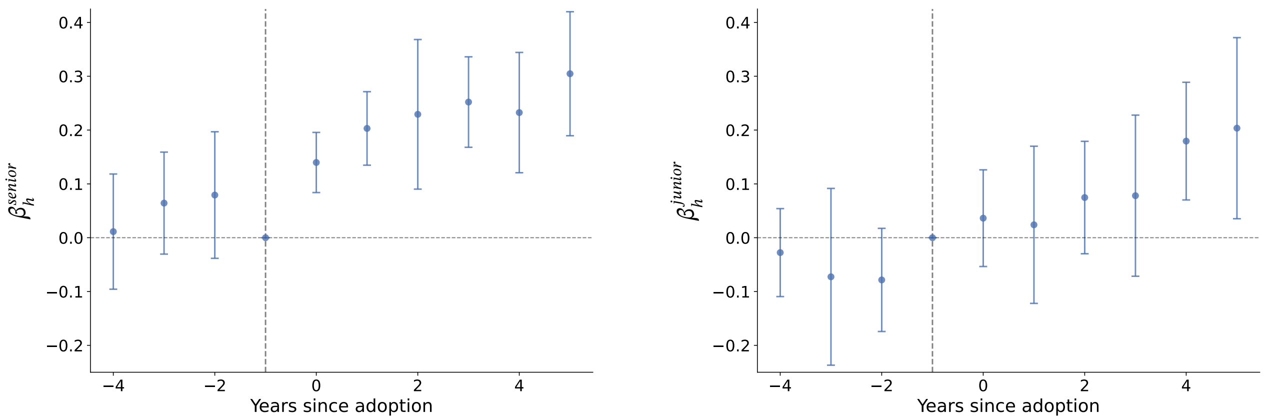
Hiring increases concentrated for senior workers. Next, I examine for which types of workers the associations are strongest. I estimate (2) for two subsamples in the data: (1) letting c be the set of all senior workers defined as workers with job titles corresponding to seniority levels of managers, directors and executives and (2) letting c be the set of all junior workers. Estimates of coefficients are reported in Figure 2. In the data, senior hires jumps up immediately at the time of adoption while the coefficients for junior hires only become significant after year 4. Again, estimates are robust to excluding the first AI workers within a firm (see Figure C.6).

This is consistent with the view that previous experience accumulated over time and practical application of AI technology is most important in the early phases of adoption and more difficult to access through more junior hires. I view this as indicative of the view that firms use poaching to adopt a technology and not simply to access technical skills after successfully adopting AI. But as AI use matures inside a firm, technical skills may become the primary constraining factor and junior hires increase.[13]

3 Model

How does a radical new technology diffuse across an economy, and how is this process different from incremental productivity improvements? In this section, I propose an equilibrium model that studies the

[13] These results are consistent with the view that AI adoption in general leads to teams becoming more junior (Babina et al., 2024), but that senior workers are necessary to form such AI teams at the start.



(a) Senior.

(b) Junior.

Figure 2: Excess probabilities by seniority status.

Notes: This graph plots the coefficient estimates β_h^c from (2) and corresponds to how excess probabilities $\gamma_{cj\ell t+h}$ change with distance to the adoption year t, separately for two subgroups c: senior workers (e.g. managers and directors) and junior workers (e.g. analysts). The left panel shows coefficients for the senior subsample. The right panel shows coefficients for the junior subsample. Estimated on subsample of E-E movers at annual frequency. The regression controls for industry-year and location fixed effects as well as pre-adoption variables related to selection into adoption (e.g. sales per worker). The sample period is 2010-2024. Destination firms in the sample are Compustat firms. Origin firms in the sample is the universe of firms in Revelio Labs. Standard errors are clustered at the firm-year level.

dynamics of an economy after an introduction of a new technology in a setting of incremental firm and worker learning and embedded in a standard job ladder model with on the job search and knowledge spillovers.

Environment. Time is continuous and runs forever. At any time $t \in \mathbb{R}$, the economy is endowed with a set $\mathcal{T}_t$ of technology vintages. Prior to $t = 0$ there exists a single technology vintage, so $\mathcal{T}_t = \{0\}$. At $t = 0$, a new vintage becomes available exogenously, thereafter $\mathcal{T}_t = \{0, 1\}$ for all $t \geq 0$. The arrival of the new technology is unanticipated prior to $t = 0$, and there are no other aggregate shocks.

The economy is populated by two types of agents: a unit mass of workers and a mass M of entrepreneurs. Agents are risk neutral and have perfect foresight over the equilibrium paths of aggregate variables.

3.1 Workers

Workers are infinitely lived and discount the future at rate ρ. Their intertemporal utility is linear and workers maximize the net present value of discounted utility over consumption. At any time t a worker is indexed by three states: (i) the most recent technology vintage in which the worker has experience $\tau \in \{0, 1\}$ (ii) the vintage-specific human capital $h \in \mathbb{R}^+$ and (iii) their employment status. The mapping of human capital between technology vintages is detailed below.

Employed workers inelastically supply one unit of labor to entrepreneurs and receive an endogenous wage w which is determined following the sequential auction bargaining protocol by Postel-Vinay and Robin (2002) where entrepreneurs Bertrand compete for the worker. Unemployed workers receive unemployment benefits bh. As described in detail below, employed workers accumulate human capital specific to their vintage through on-the-job learning. When unemployed, their human capital depreciates at rate α_u.

3.2 Entrepreneurs

Entrepreneurs can either be matched with a single worker or operate solo. Matched entrepreneurs produce output and decide whether to upgrade their technology vintage. Solo entrepreneurs produce output and post vacancies to hire a worker either out of unemployment or poach an employed worker, but cannot adopt a new vintage without a worker who has experience with the new technology. Both types of entrepreneurs discount the future at the real interest rate r_t.

3.2.1 Matched entrepreneur

The state of a matched entrepreneur is given by the tuple (z, h, τ), where $z \in \mathbb{R}^+$ is the entrepreneur's idiosyncratic, vintage-specific productivity, h the worker's human capital, and $\tau \in \{0, 1\}$ the technology vintage that is used in operation. The production process combines z, h and τ into net output $\pi(z, h \mid \tau) = A^\tau \left(z^\zeta + h^\zeta\right)^{1/\zeta} - \tau k_f$. The production function features complementarities in production between worker and firm knowledge whenever it is supermodular, i.e. $\zeta < 1$, regardless of technology vintage. The new technology vintage, i.e. $\tau = 1$, modifies the production process in two ways. First, it scales output by a factor $A > 1$. Second, it introduces a per period fixed cost $k_f > 0$. This captures the idea that modern technologies which heavily rely on intangible investments tend to lower marginal costs while increasing fixed costs in production (De Ridder, 2024).[14] A match is exogenously separated at rate κ at which point the worker becomes unemployed and the entrepreneur solo.

Learning dynamics. During a match, firm and worker knowledge evolve endogenously as a two-sided process. Following the "learning from coworker" literature, a worker learns from the entrepreneur at a rate proportional to the knowledge gap, $\alpha\,[z-h]^+$, where $x^+ = \max\{x, 0\}$. At the same time, worker knowledge gets embodied in the firm at a rate $\alpha\,[h-z]^+$. For example, workers may write manuals transcribing their knowledge or pass on tacit knowledge about ways in which a technology can or – often more importantly – cannot be operationalized. The simple process outlined here is a stand-in for all these forms of within match knowledge spillovers. Agents only learn from a more knowledgeable peer and cannot unlearn through low quality matches. This assumption is motivated by empirical evidence from the learning from coworkers literature (Herkenhoff et al., 2024; Jarosch et al., 2021), which find that workers predominantly learn from more knowledgeable peers. In addition, firm productivity is subject to exogenous idiosyncratic shocks, which capture all other shocks to a match's profitability (e.g. management skills, business opportunities, idiosyncratic demand shocks etc.).

Taken together, the law of motions of z and h within a match are given by

$$\begin{aligned} dz_t &= \alpha\,[h_t - z_t]^+\,dt + \vartheta\,(\mu - \log z_t)\,z_t dt + \sigma z_t dB_t \\ dh_t &= \alpha\,[z_t - h_t]^+\,. \end{aligned} \tag{3}$$

The first term of the first line captures the drift adjustment due to the knowledge spillovers, while the second term captures the mean-reverting nature of the exogenous shocks and dB_t denotes a standard Brownian motion. Combined, the stochastic process for the entrepreneur's productivity is a geometric Ornstein-Uhlenbeck process that incorporates knowledge spillover.[15] The second line captures the on the

[14] In the present model, the fixed cost ensures that it is the most productive entrepreneurs that first adopt the new technology, consistent with the data (Hampole et al., 2025).

[15] The SDE can equivalently be written as $dz_t = \vartheta\,(\hat{\mu}_t - \log z_t)\,z_t dt + \sigma z_t dB_t$ with $\hat{\mu}_t = (\alpha/\vartheta)\,[h_t/z_t - 1]^+ + \mu$, which itself

job learning by workers. If the entrepreneur is more knowledgeable, the worker's human capital catches up at a rate proportional to the knowledge gap. There are no shocks to worker's human capital directly, though human capital inherits the stochastic nature through its coupling with the entrepreneur's productivity.

A salient feature of (3) is that the stochastic process of human capital is asymmetric in the following sense. A series of sufficiently many positive shocks to the entrepreneur's productivity leading to $z > h$ induces positive human capital growth, whereas a series of sufficiently many negative shocks leading to $z < h$ does not affect human capital. This captures the process of volatility harvest, during which the upside of volatility is captured while protecting against downside risk of negative shocks. This process delivers positive trend growth as long as a match persists. As shown in Appendix A.6, if a match were to persist forever, then z and h would grow unboundedly. This captures the process of innovation or learning about a technology as in Atkeson and Kehoe (2007) in a reduced form way. In total, the learning process generates returns to match duration.

Technology adoption. Beyond incremental learning within an existing technology vintage, worker-entrepreneur pairs can undertake discrete technology upgrades. At the start of each period, the match jointly decides whether to adopt a new technology vintage. Upgrading to a new technology vintage entails the loss of vintage-specific expertise of both entrepreneur and workers: there is incomplete and stochastic transfer between knowledge of operating the old and the new technology vintage. Specifically, upon agreeing on using technology vintage τ with initial knowledge states (z_{t-}, τ_f) and (h_{t-}, τ_w) for entrepreneur and worker, respectively, their productivity and human capital in the new vintage become $z_t = \delta^{\tau(1-\tau_f)} z_{t-}$

follows a time-varying stochastic process. Learning endogenizes the long-run mean of the stochastic productivity process of entrepreneurs.

and $h_t = \delta^{\tau(1-\tau_w)} h_{t-}$.[16] The knowledge transfer parameter δ follows the two-state process[17]

$$\delta = \begin{cases} \overline{\delta} & \text{w.p. } \mathcal{P}(\tau_f, \tau_w) \\ \underline{\delta} & \text{otherwise.} \end{cases} \tag{4}$$

The mapping of knowledge between vintages has the following key properties. First, knowledge is fully transferable for the agent that has used the technology previously. For example, suppose the worker has prior experience (i.e. $\tau_w = 1$) with the technology and the entrepreneur does not (i.e. $\tau_f = 0$), then knowledge with the new technology ($\tau = 1$) is $z_t = \delta z_{t-}$ and $h_t = h_{t-}$. In this sense, it is cheaper for an entrepreneur to adopt the new vintage when hiring a worker with prior experience. Second, there are direct knowledge spillovers from adopters to non-adopters captured by the stochastic process and the probability mapping $\mathcal{P}(\tau_f, \tau_w)$ if $\min\{\mathcal{P}(1,0), \mathcal{P}(0,1)\} > \mathcal{P}(0,0)$. This captures the fact that experience with the new technology can help best apply the old knowledge to the new tasks and therefore improves the chances of greater transferability.[18] At last, knowledge is fully transferable when downgrading to an older technology vintage (i.e. $\tau = 0$).[19]

3.2.2 Solo entrepreneur

The state of a solo entrepreneur is given by the tuple (z, τ), where z and τ are defined analogously to above. Production occurs without a worker, so net output is $\pi(z, 0 \mid \tau) = A^\tau z - \tau k_f$. Productivity is pinned down exogenously by the mean-reverting Ornstein-Uhlenbeck process. There is no opportunity to upgrade the technology vintage without a worker. Solo entrepreneurs face an additional exogenous risk of exiting d in which case they get a value of zero and are replaced by identical solo entrepreneurs. This increases the

[16]This is written in parsimonious form. If the match decides to adopt, then $\tau = 1$. If the worker has prior experience in the new technology, then $\tau_w = 1$. If the firm has prior experience in the new technology, then $\tau_f = 1$. Taken together, limited transferability only arises whenever (i) the match is upgrading to $\tau = 1$ and (ii) agent $i \in \{w, f\}$ has no prior experience, i.e. $\tau_i = 0$.

[17]Instead of assuming a two-state stochastic process we could also assume that the transferability parameter depends on τ_f and τ_w, i.e. $\delta(\tau_f, \tau_w)$. This delivers qualitatively the same results, but is slightly more parsimonious.

[18]An alternative interpretation is that prior experience helps with avoiding crucial mistakes in the beginning.

[19]Downgrading may occur whenever worker and entrepreneur do not have the same technology state, i.e. $\tau_f \neq \tau_w$. However, I assume that if worker and entrepreneur both have used the new technology vintage, i.e. $\tau_f = \tau_w = 1$, then they do not downgrade their technology – it is an absorbing state. In the calibrated model this assumption is non-restrictive.

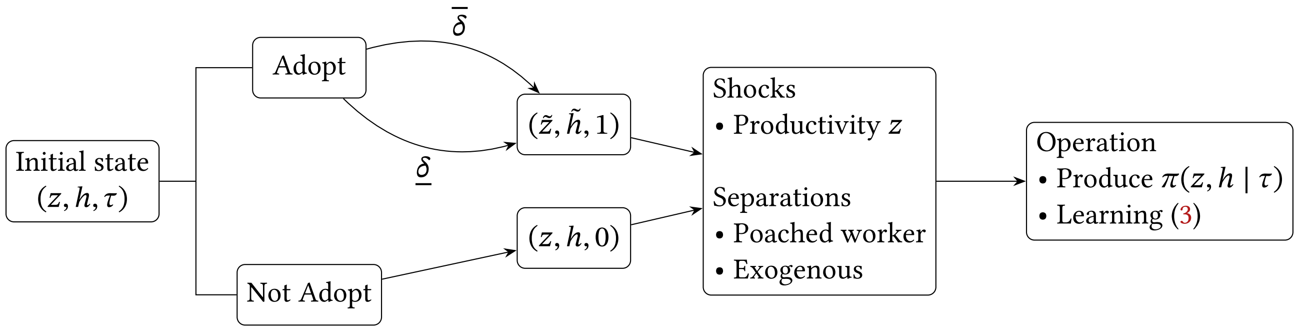


Figure 3: Decision tree for hiring and technology adoption by matched entrepreneur.

Notes: This timeline shows the sequence of events for a matched entrepreneur. The period is entered with an initial state (z, h, τ). The match first decides whether to adopt the new technology vintage. If adopt, the match is subject to the stochastic transferability parameter. The new state variables realize and the rest of the period unfolds: productivity shocks, exogenous and endogenous separations alongside production and learning.

effective discount rate of solo entrepreneurs to $r_t + d$ and captures that letting go of the worker incurs some additional risk.

A solo entrepreneur posts vacancies v at a convex cost $c(v) = \bar{c}/(1+\gamma)v^{1+\gamma}$ with $\gamma > 0$ which generates meetings with a worker at rate $q_t \cdot v$. Entrepreneurs take the aggregate contact rate q_t as given but choose the number of vacancies v to maximize expected profits. Upon meeting, entrepreneur and worker enter the sequential auction bargaining process by Postel-Vinay and Robin (2002). If a new match is formed, entrepreneur and worker optimally choose the technology vintage based on their current knowledge (z, τ_f) and (h, τ_w). The transferability of knowledge from old to new vintage takes the same form as in (4). The sequence of events is summarized in Figure 4.

3.3 Matching technology and bargaining

Solo entrepreneurs and workers meet in a frictional labor market. Search is random. Meetings are mediated through a constant returns to scale matching function $m(\mathbb{X}_t, \mathbb{V}_t) = \chi \mathbb{X}_t^{1-\beta} \mathbb{V}_t^{\beta}$, where $\mathbb{X}_t = u_t + \xi(1 - u_t)$ denotes the effective pool of job seekers and $\mathbb{V}_t$ the aggregate vacancies. The aggregate unemployment rate is denoted by u_t and ξ the relative search intensity of employed workers. Labor market tightness is denoted by $\theta_t = \mathbb{V}_t/\mathbb{X}_t$ and $p(\theta_t) = m(\mathbb{X}_t, \mathbb{V}_t)/\mathbb{X}_t = \chi \theta_t^{\beta}$ and $q(\theta_t) = p(\theta_t)/\theta_t$. An unemployed worker

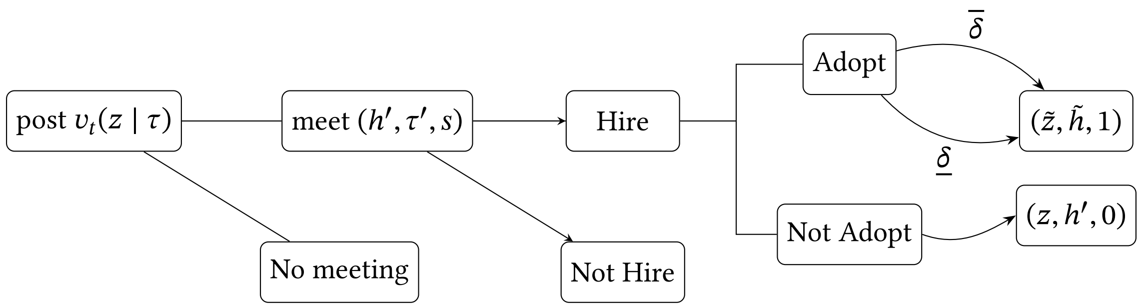


Figure 4: Decision tree for hiring and technology adoption.

Notes: This timelines shows the sequence of events of a solo entrepreneur posting vacancies and searching for a worker. A worker's employment state is denoted by $s \in \{U, E\}$ where employment state E is indexed by entrepreneur z' who the worker is matched to — which is important for the worker's outside option in bargaining. If a worker is hired, the new match decides on which technology vintage to use subject to limited transferability of prior knowledge. At the end, worker and entrepreneur agree on the same technology vintage.

is contacted at a rate $q(\theta_t)$ per vacancy posted while an employed worker gets contacted at a rate $\xi q(\theta_t)$. Likewise, an unemployed worker meets a solo entrepreneur at a rate $p(\theta_t)$, while an employed worker meets a solo entrepreneur at a rate $\xi p(\theta_t)$.

Match surplus is divided between the entrepreneur and the worker following the sequential auction framework developed by Postel-Vinay and Robin (2002). Entrepreneurs Bertrand-compete through a sequence of wage bidding contests. That is, an employed worker receives an outside offer, the incumbent employer has the option to counter-offer, and the winner is the firm that has the higher match surplus with the worker consistent with the allocation in a first price ascending auction. The winning bidder pays just enough to leave the losing firm indifferent between matching the offer and retaining and losing the worker. When an unemployed worker receives an outside offer, the entrepreneur hires the worker if the match generates a positive surplus, in which case they promise a value to the worker that leaves them indifferent between working and remaining unemployed. Wages adjust to implement the allocation, which is feasible since utility is transferable. Overall, the bargaining protocol implies that worker mobility is bilaterally efficient.

3.4 Recursive representation

Having described the economy's environment, this section details the value functions that determine the *allocative* decisions in the economy and formalizes the environment described in the preceding section. The sequential auction protocol together with transferable utility implies that allocative decisions do not depend on the split of the value of a match between worker and entrepreneur, but only on its total value.[20] As utility is linear over consumption, the real interest rate is pinned down by the discount rate, i.e. $r_t = \rho$ for all t.

Let the total value of a match between an entrepreneur and a worker be denoted by $S_t(z, h \mid \tau)$, the value of a solo entrepreneur by $V_t(z \mid \tau)$, and the value of an unemployed worker by $U_t(h \mid \tau)$. The total value of a job $S_t(z, h, \tau)$ satisfies the following HJB[21]

$$
\begin{aligned}
\rho S_t(z,h|\tau) - \partial_t S_t(z,h \mid \tau) = {} & \underbrace{\pi(z,h|\tau)}_{\text{flow surplus}} + \underbrace{\partial_z S_t(z,h|\tau)\alpha\,[h-z]^+ + \partial_h S_t(z,h|\tau)\alpha\,[z-h]^+}_{\text{learning gain}} \\
& + \underbrace{\partial_z S_t(z,h|\tau)\vartheta(\mu - \log z)z + \frac{\sigma^2 z^2}{2}\partial_{zz} S_t(z,h|\tau)}_{\text{exogenous shocks}} \\
& + \underbrace{\kappa\,(V_t(z|\tau) + U_t(h|\tau) - S_t(z,h|\tau))}_{\text{separation shock}},
\end{aligned}
\tag{5}
$$

subject to the boundary conditions for adoption and endogenous separation

$$
S_t(z,h|\tau) \geq S_t^A(z,h \mid \tau,\tau), \qquad \text{and} \qquad S_t(z,h|\tau) \geq V_t(z|\tau) + U_t(h|\tau),
$$

where the value of adopting is given by $S_t^A(z,h \mid \tau_f, \tau_w) \equiv \mathbb{E}\left[S_t(\tilde{z}, \tilde{h} \mid 1) \mid z, h, \tau_f, \tau_w\right]$, where $\tilde{z} = \delta^{1-\tau_f} z$ and $\tilde{h} = \delta^{1-\tau_w} h$ and δ follows the stochastic process defined in (4).

Let the endogenous cumulative distribution function of active matches, and unemployed be given

[20]Technically, we must also assume that adoption decisions are made jointly by worker and entrepreneur in order to maximize the total value. Bilal et al. (2022) show how this can be rationalized as the outcome of a bargaining process. Intuitively, if there are actions that could increase the total value of a job there should exist a bargaining protocol that apportions the surplus.

[21]For notational convenience, we define the differential operators $\partial_z = \frac{\partial}{\partial z}$, $\partial_{zz} = \frac{\partial^2}{\partial z^2}$ and $\partial_h = \frac{\partial}{\partial h}$.

by $G^e_t(z, h \mid \tau)$ and $G^u_t(h \mid \tau)$, respectively. Define the value of an entrepreneur with states (z, τ) hiring a worker with states (h', τ'), taking into account the optimal choice of the technology vintage, to be $S^H_t(z, h \mid \tau_f, \tau_w) = \max\left\{S^A_t(z, h \mid \tau_f, \tau_w),\ S_t(z, h \mid 0)\right\}$. Then, the value of a solo entrepreneur is

$$
\begin{aligned}
(\rho + d)V_t(z|\tau) - \partial_t V_t(z|\tau) = \ & \underbrace{\pi(z, 0|\tau)}_{\text{flow profit}} + \underbrace{\partial_z V_t(z|\tau)\vartheta\left(\mu - \log z\right) z + \frac{\sigma^2 z^2}{2}\partial_{zz}V_t(z|\tau)}_{\text{exogenous shocks}} \\
& \underbrace{+ \max_v \left\{ vq(\theta_t)\frac{1}{\mathbb{X}_t}\left[\xi \sum_{\tau' \in \mathcal{T}_t} \int \Delta^{EE}_t(z', h', z|\tau', \tau)^+ dG^e_t(z', h'|\tau') + \sum_{\tau' \in \mathcal{T}_t} \int \Delta^{UE}_t(h', z|\tau', \tau)^+ dG^u_t(h'|\tau')\right] - c(v)\right\}}_{\text{expected gain from posting vacancies}},
\end{aligned}
\tag{6}
$$

where the expected gain from posting vacancies aggregates over the individual gains of potential matches

$$
\begin{aligned}
\Delta^{EE}_t(z', h', z \mid \tau', \tau) &= S^H_t(z, h' \mid \tau, \tau') - V_t(z|\tau) - \left(S_t(z', h' \mid \tau') - V_t(z' \mid \tau')\right) \\
\Delta^{UE}_t(h', z \mid \tau', \tau) &= S^H_t(z, h' \mid \tau, \tau') - V_t(z|\tau) - U_t(h|\tau').
\end{aligned}
\tag{7}
$$

The value of unemployment solves

$$
\rho U_t(h \mid \tau) - \partial_t U_t(h \mid \tau) = bh - \partial_h U_t(h \mid \tau)\alpha_u h. \tag{8}
$$

Given the boundary condition $U_t(0 \mid \tau) = 0$, the unique solution is $U_t(h \mid \tau) = bh/(\rho + \alpha_u)$.

Comment. Value functions such as $S_t(z, h \mid \tau)$ are time-varying because they depend on the aggregate state through the option value of becoming a solo entrepreneur. The key aggregate variables are labor market tightness θ_t and the endogenous distribution over matches and unemployed G^e_t and G^u_t, which determine the expected gains from posting vacancies. The value of a match inherits this time dependency through the option value of becoming a solo entrepreneur either through endogenous or exogenous separations. It follows that in a model without exogenous separations, i.e. $\kappa/\rho \to 0$, and no endogenous separations the total value of a match is constant over the transition towards the new ergodic steady state. Conversely, in the full model all time-variation in policy functions over the transition stem from changes in the option value.

3.5 Policy functions

Agents in the economy make a total of five decisions: number of vacancies posted v_t, discrete hiring choices $\mathbb{1}^{EE}$, $\mathbb{1}^{UE}$, discrete adoption choice $\mathbb{1}^{A}$, and discrete match separation choice $\mathbb{1}^{X}$.

Decisions of solo entrepreneur. First, solo entrepreneurs choose how many vacancies to post, taking as given the aggregate matching rate $q(\theta_t)$ and the distribution over potential matches dG_t^e and dG_t^u. The first order condition of (6) yields the optimal vacancy posting policy. It is given by

$$\bar{c}v_t(z \mid \tau)^\gamma = q(\theta_t)\frac{1}{\mathbb{X}_t}\left[\xi \sum_{\tau' \in \mathcal{T}_t} \int \Delta_t^{EE}(z', h', z|\tau', \tau)^+ dG_t^e(z', h'|\tau') + \sum_{\tau \in \mathcal{T}_t} \int \Delta_t^{UE}(h', z|\tau', \tau)^+ dG_t^u(h'|\tau')\right].$$

The LHS denotes the marginal cost of posting a vacancy. The RHS is the expected surplus gain from posting an additional vacancy, which is determined by the offer distributions of Δ^{EE} and Δ^{UE} defined in (7). Vacancy posting is decreasing in labor market tightness θ_t and increasing in the expected surplus gain averaged over the population distributions G_t^e and G_t^u. Matched workers tend to have a higher human capital which generates larger expected match values, but poaching is more expensive. On the other hand, unemployed workers have a lower outside option and entrepreneurs can extract a larger fraction of the surplus. A greater availability of workers with knowledge in the new technology vintage tend to increase the expected value of new matches, but further intensifies competition for workers.

Conditional on meeting with an employed or unemployed worker the solo entrepreneur next engages in the sequential auction to bid for the worker. As discussed, a new match is formed whenever it is bilaterally efficient to do so. From (7) this is determined by Δ_t^{EE} and Δ_t^{UE}. The hiring policies are therefore given by the indicator functions

$$\mathbb{1}^{EE}\left(z', h', z \mid \tau_w, \tau_f; t\right) = \mathbb{1}\left\{\Delta_t^{EE}(z', h', z \mid \tau_w, \tau_f) > 0\right\}$$
$$\mathbb{1}^{UE}\left(h', z \mid \tau_w, \tau_f; t\right) = \mathbb{1}\left\{\Delta_t^{UE}(h', z \mid \tau_w, \tau_f) > 0\right\}.$$

There are three primary forces that determine the hiring policies: (i) production gains and complementarities in z and h (ii) learning gains governed by the learning rate α as well as the scope of learning $|z - h|$

and (iii) the value of outside options V_t and U_t. The option to adopt a new technology vintage via hiring increases both production complementarities and learning gains. That is, entrepreneurs are more keen to poach, but also to retain workers with the new technology vintage. Details are provided in Appendix A.2, including a decomposition of these main factors.

Once a match is formed, the entrepreneur-worker pair decides which technology is optimal to use during operation. The match uses the new technology vintage whenever

$$\mathbb{1}^A\left(z, h \mid \tau_f, \tau_w; t\right) = \mathbb{1}\left\{S_t^A(z, h \mid \tau_f, \tau_w) \geq S_t(z, h \mid 0)\right\}.$$

The combination of operational fixed cost and greater productivity with the modern technology implies that there exists a cut-off productivity $\underline{z}_t(h)$ such that for all $z \geq \underline{z}_t(h)$ adoption occurs and no adoption otherwise.[22] This is a quantitative result in line with empirical results around AI adoption, documenting that larger, high sales, and more productive firms are more likely to adopt (Hampole et al., 2025).

Separation decision of matched entrepreneur. At the beginning of each period matched entrepreneurs decide whether to terminate the match $\mathbb{1}^X(z, h \mid \tau; t) = \mathbb{1}^X\{S_t(z, h \mid \tau) \leq V_t(z \mid \tau) + U_t(h \mid \tau)\}$.

3.6 Equilibrium

The economy before the arrival of the new technology vintage is in its ergodic steady state for $t < 0$. The stationary equilibrium is defined as follows.

Definition 1 (Stationary equilibrium)**.** Given $\mathcal{T}_t = \{0\}, \forall t < 0$ the initial stationary equilibrium prior to $t = 0$ is defined as the set of value functions $\{S_{ss}, V_{ss}, U_{ss}\}$, distribution functions $\{G^e_{ss}, G^v_{ss}, G^u_{ss}\}$, policy functions $\{v_{ss}, \mathbb{1}^{EE}_{ss}, \mathbb{1}^{UE}_{ss}, \mathbb{1}^{A}_{ss}, \mathbb{1}^{X}_{ss}\}$, labor market tightness θ_{ss} such that (5), (6), and (8) hold, labor market tightness is constant and consistent with aggregate vacancies $\mathbb{V}_{ss} = \int v_{ss}(z \mid 0) dG^v_{ss}(z \mid 0)$ and aggregate search input $\mathbb{X}_{ss} = \int dG^u_{ss}(h \mid 0) + \xi \int dG^e_{ss}(z, h \mid 0)$, and distributional dynamics are constant and consistent with policies and worker flows.

[22] Technically, there is an opposite force coming from the assumption on limited transferability of knowledge. For example, lower productivity entrepreneurs have less to lose upon switching to the new technology. In the calibration used in the main exercise this effect is confirmed to not be large and to not dominate.

After the introduction of the new technology vintage, the economy transitions to its new steady state for $t \geq 0$. The equilibrium over the transition is defined as follows.

Definition 2 (Transition path)**.** Given the initial distribution over matches, solo entrepreneurs, and unemployed $\{G^e_{ss}, G^v_{ss}, G^u_{ss}\}$ an equilibrium over the transition after the unanticipated arrival of the new technology vintage is characterized by the full sequence of value functions $\{S_t, V_t, U_t\}$, distribution functions $\{G^e_t, G^v_t, G^u_t\}$, policy functions $\{v_t, \mathbb{1}^{EE}, \mathbb{1}^{UE}, \mathbb{1}^{A}, \mathbb{1}^{X}\}$, and labor market tightness $\{\theta_t\}$ such that (5), (6), and (8) hold, labor market tightness is consistent with aggregate vacancies $\mathbb{V}_t = \sum_{\tau \in \mathcal{T}_t} \int v_t(z \mid \tau) dG^v_t(z \mid \tau)$ and aggregate search input $\mathbb{X}_t = \sum_{\tau \in \mathcal{T}_t} \int dG^u_t(h \mid \tau) + \xi \sum_{\tau \in \mathcal{T}_t} \int dG^e_t(z, h \mid \tau)$, and distributional dynamics are consistent with policies and worker flows. A complete characterization of the distributional dynamics using the Kolmogorov Forward Equation is provided in Appendix A.1.

3.7 Properties of the Model

This section discusses key definitions and model properties.

Labor Mobility. Worker mobility serves three distinct functions. First, reallocation facilitates assortative matching based on human capital and firm productivity, generating output gains through complementarities. Second, worker transitions enable knowledge diffusion by generating matches with positive learning gains. Third, labor flows facilitate technology adoption through the transfer of vintage-specific knowledge embodied in workers and firms and reducing the effective cost of adoption.

Knowledge Diffusion. This is the act of transfer of knowledge between two agents. I distinguish between two types. *Gradual diffusion* occurs within ongoing employment relationships through incremental learning. *Discrete diffusion* occurs at match formation when technology adoption decisions are made, generating discontinuous changes in knowledge and technology. The former benefits more strongly from match duration while the latter occurs at a distinct point in time.

Channels of Adoption. Entrepreneurs and workers can adopt the frontier technology vintage either internally or externally. I distinguish between the following channels.

Definition 3. Entrepreneurs adopt the new technology through

1. *Internal adoption*: Existing matches upgrade to the new vintage subject to the incomplete transferability of their prior knowledge.
2. *Adoption via hiring*: Entrepreneurs without the modern technology hire experienced workers and adopt the new vintage upon forming the new match. This channel is further divided into poaching and hiring out of unemployment.

Definition 4. Workers adopt the new technology through

1. *Internal adoption*: Existing matches upgrade to the new vintage subject to the incomplete transferability of their prior knowledge .
2. *Adoption via training*: Experienced entrepreneurs hire inexperienced workers and use the modern technology in the new match. This channel is further divided into poaching and hiring out of unemployment.

Example paths. Figure 5 plots illustrative example paths and demonstrates key dynamics of the model. In the illustrations, there are two entrepreneurs, A and B, and a single worker. Figure 5a depicts dynamics relating to learning and poaching within a given vintage. Figure 5b depicts dynamics relating to adoption and poaching. Figure 5c depicts output dynamics under adoption and non-adoption scenarios.

Learning and poaching within vintage. Initially, the worker is matched with entrepreneur A while entrepreneur B operates solo. Worker and entrepreneur A have identical knowledge levels z_A and h at the start. At random times, entrepreneur A experiences exogenous shocks to their productivity. A gap between the productivity of the entrepreneur A and the worker's human capital emerges, generating learning gains. The worker then incrementally learns from the entrepreneur and sees their human capital increase. Further, as long as learning gains are present the value of maintaining the match is relatively large. As the gap closes over time, gains from continuing the match fall, making it more likely for the worker to get poached. At a random time, entrepreneur B meets with the worker. If entrepreneur B has sufficiently large learning gains from hiring the worker, the worker moves from A to B even if the latter has a lower productivity. Subsequently, the worker's human capital remains constant and worker knowledge transfers to entrepreneur B, whose productivity z_B increases gradually.

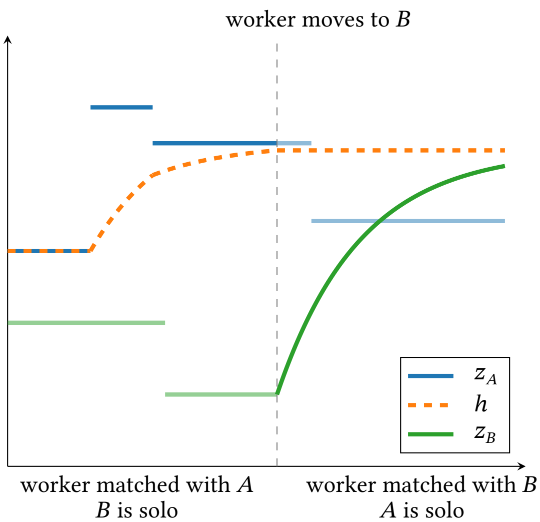
Adoption dynamics. Initially, the worker is matched with entrepreneur A. If productivity z_A and human capital h are sufficiently large, the match finds it optimal to adopt the new technology vintage. In this event, only some of the knowledge in the old technology is transferable to the new vintage. This is captured by a discrete fall in z_A and h upon adoption. In the illustration, the match draws the low transferability state and the respective knowledge in the new vintage are $\underline{\delta} z_A(0)$ and $\underline{\delta} h(0)$. Over time, as the entrepreneur receives exogenous shocks to their productivity z_A and h grow together.[23] When productivity falls below the worker's human capital, learning dynamics imply that the entrepreneur's productivity will catch up to the worker's human capital over time. This gives rise to a volatility harvesting phenomenon, implying that z_A and h grow together. This positive trend growth can be more generally interpreted as incremental follow-up improvements of an existing technology vintage or a reduced form innovation process.

At a random time, entrepreneur B meets with the worker. If it is profitable, they will form a new match and if optimal entrepreneur B will adopt the new technology in the new match. Since entrepreneur B has not used the technology previously, only a random fraction δ of the old knowledge will be transferred to the new vintage.[24] However, since the worker is already experienced in the new technology vintage entrepreneur B can immediately learn from the worker and their productivity level z_B grows even in absence of any productivity shocks. This illustrates the nature of knowledge spillovers in the model that make it relatively cheaper to adopt via external hiring than internally.

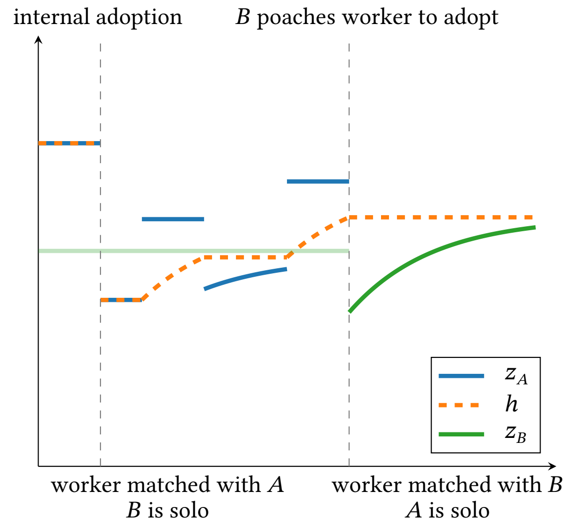
Output dynamics. The solid path depicts output if the match adopts the new technology vintage while the dashed path depicts output if the match keeps the old technology vintage. Both lines are subject to the same exogenous shocks. Upon adoption output falls initially due to the limited transferability of knowledge between vintages. Since productivity in the new technology vintage is scaled by a factor A,

[23] In the full model, this process can also be driven by the mean-reversion component of the exogenous stochastic process for productivity if z_A falls below the long-run mean. This captures the notion that learning may be easier the lower the current knowledge level. Conversely, the more productive the entrepreneur the harder it becomes to attain additional knowledge improvements.

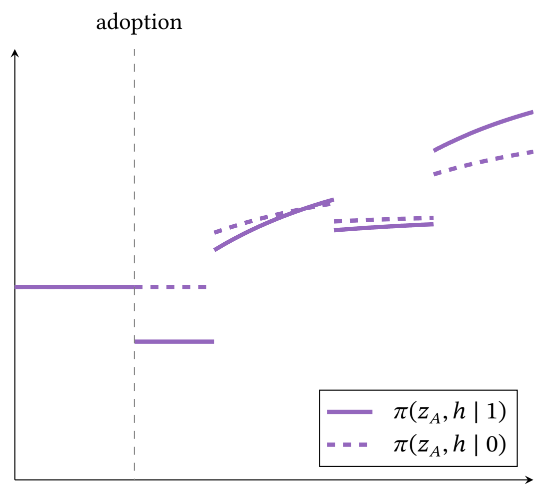
[24] In the illustration, the match draws the high transferability state, capturing the assumption that $\mathcal{P}(0,1) > \mathcal{P}(0,0)$ which makes this event more likely when adopting with an experienced worker. This assumption is not strictly necessary, but will be quantitatively important to match the observed poaching patterns in the data.



(a) Poaching and learning dynamics.



(b) Adoption dynamics.



(c) Output under adoption and non-adoption.

Figure 5: Example paths.

Notes: The top left panel shows productivity paths for two entrepreneurs A and B and the human capital path for a single worker. The worker is initially matched to A. The panel shows the learning dynamics within an existing match and when switching employer. The right panel shows productivity paths and the human capital path. The panel illustrates the limited transferability assumption when upgrading to a new technology vintage and the advantage as a technology laggard (here entrepreneur B) of poaching an experienced worker to adopt. The bottom right panel shows the output paths for a single matched entrepreneur A if they adopt (solid) and if they do not adopt (dashed). The match is subject to identical sequence of shocks, but adoption comes with limited transferability of prior knowledge.

the positive shocks to productivity improve output by a larger amount under the adoption scenario and with time output exceeds the non-adoption counterfactual. This illustrates the dynamics of lower initial output but long-run gains of employing the new technology vintage.

3.8 Calibration

This section briefly discusses the calibration. Appendix B and Table B.1 provide further details.

Initial steady state. Labor market parameters are chosen to be consistent with key labor market statistics from the U.S. In the calibrated initial ergodic steady state the unemployment rate equals 5%, quarterly employment-to-employment transition rate equals 7.8%, the quarterly job finding rate of unemployed is 72% and the quarterly job separation rate equals 4%. The calibrated productivity process produces a 90-10 differential in TFP of 0.83, which is close to but slightly above the figure stated for manufacturing firms in the U.S. in Syverson (2011), reflecting the observation that dispersion is likely higher in professional services and ICT sectors than in manufacturing. Learning dynamics parameters α and α_u are calibrated to be consistent with Herkenhoff et al. (2024).[25] The discount rate is 8% annually. The value is higher than usual to capture some degree of risk aversion, which is in line with Herkenhoff et al. (2024).

Technology parameters. A crucial factor determining the diffusion of the new technology vintage is the ease at which non-adopter are able to poach a worker from a previous adopter. In the empirical section, I found that flows from previous adopters to newly adopting firms are non-negligible. In the model these poaching flows are modulated in part by the probabilities $\mathcal{P}$. I choose them to match our empirical target. Appendix B discusses the strategy of how this is done.

4 Quantitative Model

This Section presents the main exercise. Prior to $t < 0$, the economy is at the stationary equilibrium featuring a single vintage, i.e. $\mathcal{T}_t = \{0\}$. At $t = 0$, there is an unanticipated introduction of the new

[25] The steady state version of my model with $\mathcal{T} = \{0\}$ is closely related to Herkenhoff et al. (2024).

technology vintage and $\mathcal{T}_t = \{0, 1\}$ thereafter. I now study the transition to the new ergodic steady state.

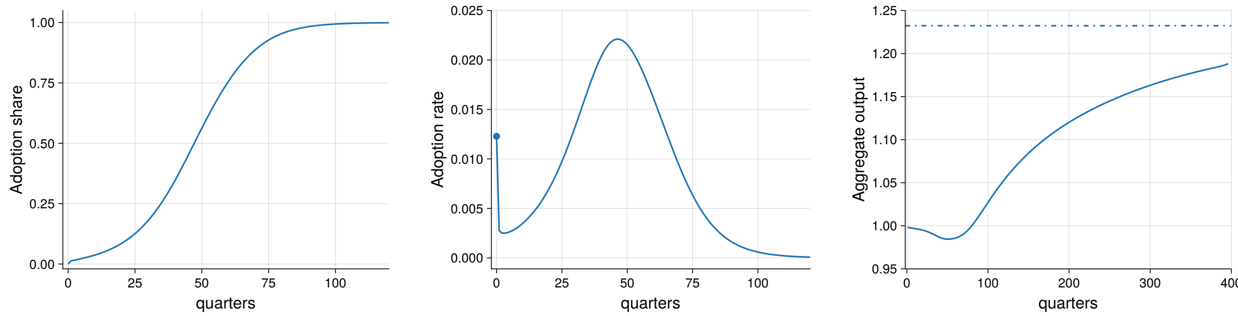
4.1 Diffusion curve.

The diffusion of the new, path-breaking technology proceeds in three phases. In the initial phase, approximately 1.5% of entrepreneurs immediately adopt the innovation upon its introduction. The presence of the per-period fixed cost implies that the most productive matches adopt first, in line with empirical evidence (Hampole et al., 2025). During the second phase, adoption rates gradually accelerate over the subsequent decade, reaching 50% penetration within 46 quarters. During the final phase, the remaining firms adopt, with complete diffusion taking approximately three decades.

Figure 6 illustrates these diffusion dynamics across three dimensions. The left panel depicts a canonical S-shaped adoption curve, demonstrating a slow initial uptake, rapid middle-phase acceleration, and eventual saturation. The center panel presents quarterly adoption rates, showing the underlying changes in technology uptake, with adoption rates peaking after 47 quarters, and the initial jump in adopters after the unexpected arrival of the new technology.[26] The right panel plots the corresponding path for aggregate output over the transition period.

The S-shaped diffusion pattern emerges endogenously from the model's random search framework when access to skilled labor is a binding constraint as widely documented in survey data in the case of AI (European Commission, 2020; Rammer & Schubert, 2022). The key mechanism behind the strong complementarities operates through labor market constraints: early in the transition access to skilled workers and workers with prior experience in the new vintage constitutes a binding constraint to adoption. In the random search framework, contact rates between entrepreneurs and workers are proportional to their existing population, which generates dynamic complementarities similar to epidemiological models of contagion. Early in the diffusion process, the scarcity of experienced entrepreneurs and workers constrains adoption. As a greater population share of entrepreneurs and workers successfully adopts the modern vintage, the likelihood of non-adopters matching with adopters experienced in the new technology increases, thereby progressively relaxing the human capital constraint and accelerating diffusion.

[26] Due to the fixed cost of operation only the most productive matches adopt initially. That is, the mass of matches above the critical adoption threshold will adopt immediately — causing a jump in the distribution.



(a) Diffusion curve **(b)** Adoption rate **(c)** Output

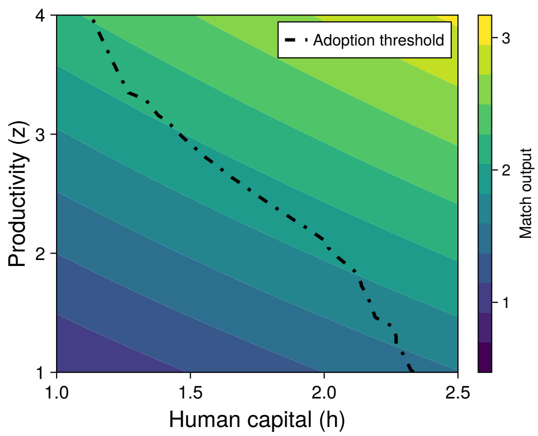
Figure 6: Adoption patterns for entrepreneurs over the transition.

Notes: The left panel plots the share of entrepreneurs that have adopted the new technology after its introduction. The middle panel plots the adoption rate, i.e. the quarterly increase in the number of entrepreneurs. The right panel plots aggregate output over the transition, taking into account changes to knowledge variables, vacancy posting and fixed costs. Time $t = 0$ denotes the time the new technology vintage is introduced.

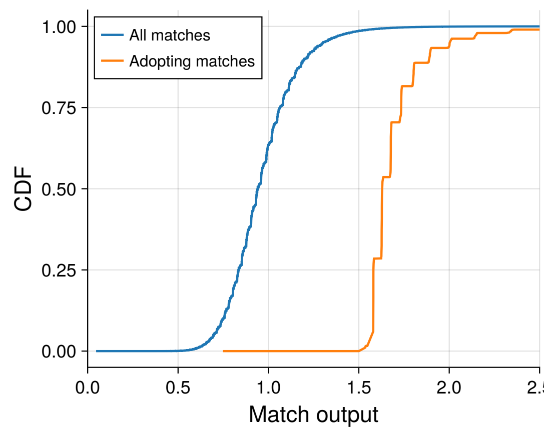
A key feature of Figure 6c is that aggregate output dynamics lag substantially behind technology adoption rates. The path for aggregate output exhibits a so-called J-curve which has been documented in the literature on intangible intensive technologies and general purpose technologies (Brynjolfsson et al., 2021; David, 1990). Initially, adopting firms experience have limited experience with the new technologies due to the partial transferability of their knowledge, and output growth occurs gradually as worker-firm matches mature and mutual learning generates incremental improvements in the productivity of entrepreneurs and the human capital of workers.[27] This pattern also aligns with theoretical frameworks emphasizing the important role of follow-up innovations after the introduction of a path-breaking technology (Helpman & Trajtenberg, 1994, 1996). The dynamics are further modulated by worker mobility: frequent cross-poaching that drives the diffusion disrupts the match-duration dependent knowledge accumulation process. This slows down learning at the individual level. Aggregate output improvements materialize only after approximately 20 years.[28]

[27] This process can be interpreted as the incremental tinkering of follow-up innovation.

[28] A richer model with e.g. multi-worker firms might feature a shorter lag between output growth and adoption, as pioneering adopters may use their technology advantage to scale up faster and constituting a greater share of output growth. Similarly, allowing for endogenous innovation intensity would have similar effects. Further, note that the model is best viewed as stationarized around a Balanced Growth Path such that aggregate dynamics may deliver small yet positive growth rates.



(a) Adoption thresholds and match output.



(b) Distributions of match output.

Figure 7: Initial adoption impulse at $t = 0$.

Notes: The left panel shows the contour lines of the initial match output $\pi(z, h \mid 0)$ and the adoption threshold $\underline{z}_t(h)$ at time $t = 0$. Adoption occurs when $z \geq \underline{z}_t(h)$. That is, to the north-east of the dashed line. The right panel plots the conditional CDF of initial match output $\pi(z, h \mid 0)$ for (i) all matches and (ii) all adopting matches. Match output is shown relative to average output.

Patient Zero: Who sets off adoption? Figure 7 illustrates initiates the diffusion process. The left panel plots the adoption threshold at impact ($t = 0$) – above which matches adopt the new technology – alongside the contour lines for initial output $\pi(z, h \mid 0)$. The figure highlights two main insights. First, more productive matches are more likely to adopt, consistent with empirical evidence on AI adoption (Hampole et al., 2025). This relationship is further illustrated in the right panel, where the CDF of initial adopters first-order stochastically dominates that of the overall population. This correlation arises because operating the new technology entails a per-period fixed cost and knowledge in the new technology scales in prior expertise.

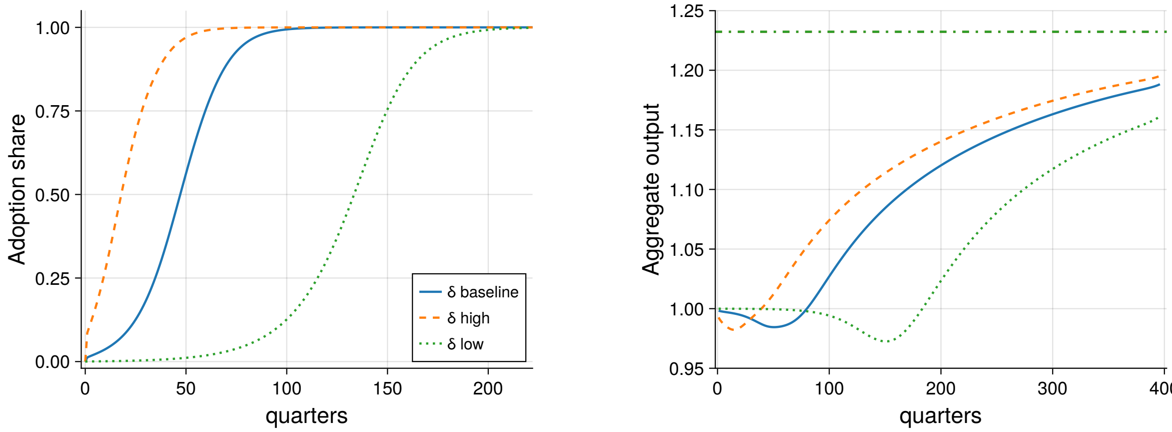
Second, the adoption threshold has a steeper gradient than the output contour lines, highlighting the role of human capital in shaping the initial adoption decision through learning gains. Higher human capital expands the scope for learning, making adoption more attractive despite its initial cost. The intuition is as follows: the more knowledgeable the worker is initially, the more knowledge they can transfer and build upon after adoption – though this transfer is only partial. Consequently, adopting the new technology with a highly skilled worker opens up greater opportunities for learning and productivity growth. Further discussion of these competing motives is provided in A.4.

Determinants of diffusion speed: Nature of radical technology. This section explores how transition dynamics differ across different types of technology. The past has seen a variety of path-breaking technologies with different set of characteristics, ranging from the steam engine in the First Industrial Revolution, electric power generation to ICT and AI. Through the lens of the framework, a crucial factor determining the speed of diffusion of each of these technologies is the degree to which entrepreneurs are constrained by access to skilled labor. This is closely linked to the degree of transferability of old skills to new technologies.[29] The more transferable old skills are to the new technology vintage the less constrained firms are internally and the lower the need for external labor to adopt the new technology. The absence of a constraint on scarce knowledge then mutes the dynamic complementarities. This can be clearly seen in the extreme case in which knowledge is perfectly transferable, i.e. $\overline{\delta} = \underline{\delta}$, — in which case there is no difference between *internal adoption* and *adoption via hiring* for a given z, h. Clearly, the role of external workers and labor markets will be muted in this case.[30]

To illustrate this dependence, Figure 8a shows the adoption paths under different knowledge transferability regimes, i.e. different values for $\underline{\delta}$ and $\overline{\delta}$. A higher transferability value, i.e. high $\mathbb{E}[\delta]$, now generates a diffusion curve that may no longer follow an S-shape, but is concave, with a large fraction of adoption occurring at impact. Firms no longer need to source talent from outside as much and adopt internally. However, adoption is still sluggish as firms require sufficiently high productivity shocks to reach the adoption threshold and currently solo entrepreneurs require a worker to adopt the new technologies, which is mediated through a frictional labor market. Conversely, a lower transferability of old skills to new vintages, i.e. small $\mathbb{E}[\delta]$, slows down diffusion as firms are relatively more constrained internally and rely on external hires to relax this constraint. The primary reason is that while internal adoption requires that both entrepreneur and worker newly learn the technology vintage, adoption via hiring only requires that the entrepreneur newly learns about the technology. Further, the assumption on bilateral learning

[29] Through the lens of the model this is equivalent to the notion of radical technology. The lower the transferability the more radical the technology.

[30] This is not to say that labor markets will have no role even in this extreme case. For example, they will be still crucial for enabling more productive firms to scale up and efficiently allocate workers across heterogeneous firms as in Hopenhayn and Rogerson (1993) and Bilal et al. (2022).



(a) Diffusion curve

(b) Output

Figure 8: Adoption patterns for different technological parameters for transferability.

Notes: The left panel plots the share of entrepreneurs that adopted the new technology over the transition for three different technological parameters for the transferability of prior knowledge. The solid line represents the calibrated baseline. The dashed line corresponds to a higher transferability technology while the dotted line corresponds to a technology with lower transferability. The right panel plots the respective paths for aggregate output.

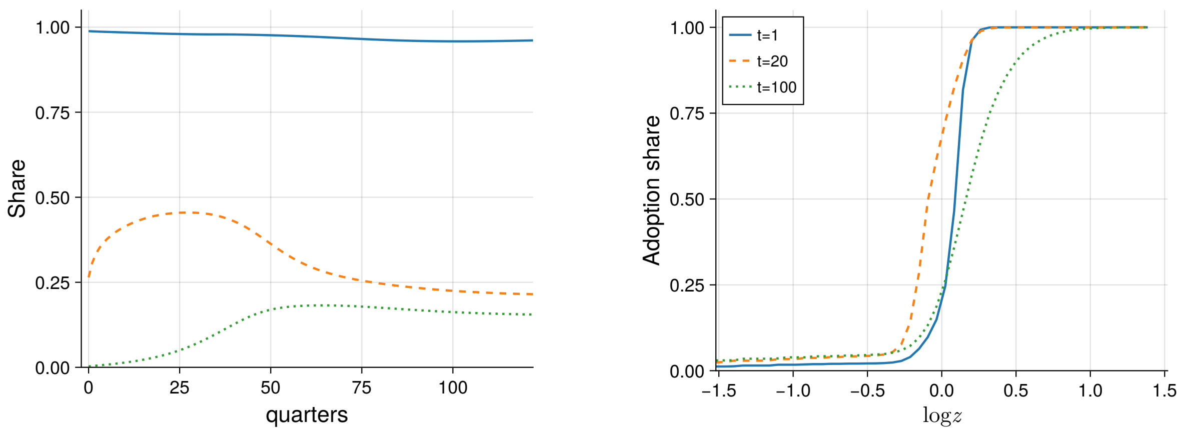
implies that part of the worker's knowledge gets incrementally transferred to the entrepreneur.

These dynamics find support in (Adão et al., 2024). Comparing the speed of the diffusion of manufacturing and ICT technologies, the authors show that the ICT revolution brought about large changes in the type of tasks being performed, while the new manufacturing jobs were largely able to reuse existing skills. As a consequence, the speed of diffusion is documented to have been faster for the latter than the former.

4.2 Importance of worker flows

Access to skilled workers as binding constraint. A key argument in the previous sections is that the speed of diffusion and shape of the diffusion curve crucially depends on the extent to which access to external skilled workers experienced in the new technology constitutes a bottleneck. This section examines and confirms this channel in the model.

In the left panel of Figure 9, I show three different measures for the desire and ability to adopt the



(a) Measures of desire and ability to adopt via poaching.

(b) Ability to poach experienced worker by entrepreneur productivity.

Figure 9: Skilled worker as a constraint to technology adoption.

Notes: The left panel plots the fraction of all non-adopter solo entrepreneurs who *would* adopt if matched with an experienced worker (blue, solid), the fraction of non-adopter solo entrepreneurs who *do* poach and adopt conditional on meeting an experienced worker (orange, dashed), and the fraction of non-adopter solo entrepreneurs who *do* poach and adopt via poaching an experienced worker. The right panel plots the fraction of non-adopter solo entrepreneurs who are able to to hire an experienced worker and adopt the new technology.

new technology. First, the solid line plots the share of solo, non-adopter entrepreneurs that *would* newly adopt conditional on meeting with a skilled worker over the transition. Nearly all entrepreneurs would adopt if they could access an appropriately skilled worker.[31]

Second, the dashed line in the left panel plots the share of solo, non-adopter entrepreneurs that *do* newly adopt conditional on meeting with a skilled worker. The difference between the solid and the dashed lines captures the (in-)ability of a solo entrepreneur to successfully poach a worker from a previously adopting firm. The share follows a hump shape over the transition. Initially, only the most productive matches adopt such that pioneering adopters are highly productive, making it hard to poach from. As the technology diffuses to less productive firms poaching becomes relatively easier. At last, as technology-specific knowledge is accumulated within matches the value of continuing a worker-entrepreneur relationship increases and poaching becomes harder again. These dynamics are also shown in the right panel of Figure

[31]The moderate fall in the share over the transition captures the compositional effects that the most productive matches will adopt first, implying that the remaining pool of non-adopters is relatively less productive towards the later stages of the transition.

9, which shows which entrepreneurs are constrained.[32] Initially, the line shifts to the left indicating that poaching becomes feasible for a larger share of entrepreneurs and then over the long-run shifts back to the right.[33]

Third, the dotted line in the left panel plots the fraction of solo, non-adopter entrepreneurs that newly adopt with a skilled worker, unconditionally. The difference between the dashed and the dotted line captures the probability of meeting with a skilled, experienced worker. The increase in the dotted line thus captures the rising probability of encountering an experienced worker over the transition. This most directly captures the notion that it is the relaxation of the constraint on skilled labor drive the diffusion curve.

Which worker flows matter? This section examines how the economy overcomes the constraint on experienced labor and specifically which type of worker flows matter at different points in the transition. To this end, I decompose the relative importance of the distinct adoption channels. Following Definition 3, I categorize adoption by entrepreneurs into two channels: (1) *internal adoption* (2) *adoption via hiring*. The latter channel is further distinguished by whether a solo entrepreneur forms the new match with a poaching hire or with an unemployed worker.[34]

Figure 10 illustrates the relative contributions of those channels over the first three decades of the transition period. The left panel plots the share of newly adopting entrepreneurs coming from each of the three channels. The right panel plots the absolute quarterly rate at which entrepreneurs newly adopt. This is a decomposition in the sense that the sum of the three lines equals 6b.[35] The importance of each channel varies over the transition.

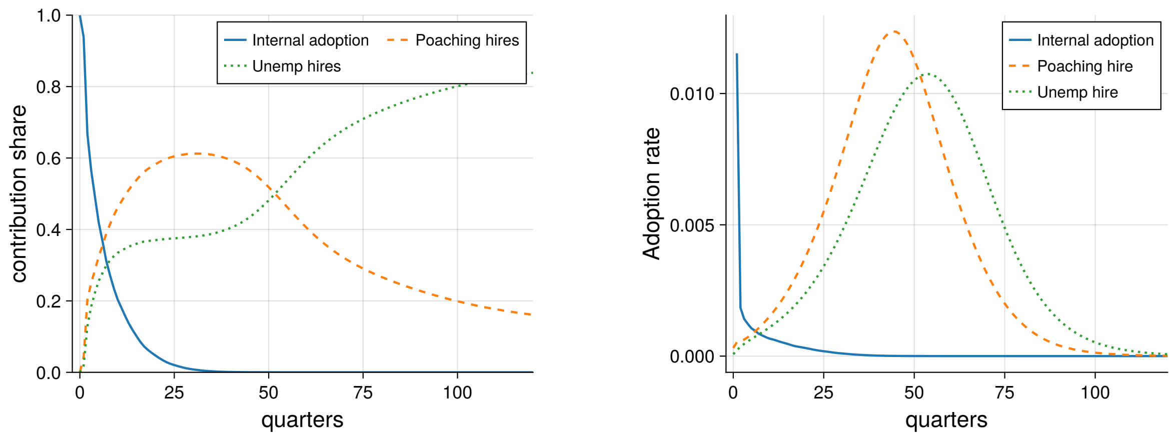
Internal adoption. Initially, internal adoption is the primary driver behind the diffusion. However, the

[32] Specifically, lower values imply a larger degree of being constrained.

[33] The lines shift upwards over the transition as matches with the new technology vintage separate over time and unemployment represents slower reallocation dynamics that are accessible even to low productivity entrepreneurs.

[34] Likewise, adoption by workers is categorized into two channels: (1) internal adoption and (2) adoption via training following Definition 4. The latter channel is again distinguished by a poaching or unemployment hire.

[35] The notion of a decomposition is subtle. Conceptually, all adopters can be traced back to internal adopters who are necessary to set the process in motion. That is, all contagion is traced back to a set of *patient zeros*. Contribution in this context refers to the breakdown of the various channels of adoption. For example, it does not take explicitly into account that adoption via poaching will by itself trigger more adoption via unemployment hires in the future.



(a) Entrepreneur adoption

(b) Marginal

Figure 10: Contribution of adoption channels

Notes: The left panel plots the share of newly adopting entrepreneurs attributable to each of three channels: internal adoption (blue, solid), poaching hires (orange, dashed), and unemployment hires (green, dotted). The right panel plots the total quarterly rate of newly adopting entrepreneurs attributable to each respective channel. Further details are provided in Appendix A.5.

importance of this channel quickly diminishes as free-riding incentives begin to dominate. That is, as the share of experienced workers increases in the population the option value to wait to adopt via hiring such a worker becomes increasingly more attractive (see Appendix D.1). This, reduces the willingness of entrepreneurs to do more costly internal adoption.

Poaching hires. During the accelerating phase of diffusion, adoption is primarily driven by poaching. The shape of this curve mirrors Figure 9a, reflecting the time-varying ease of hiring workers away from other firms.

Unemployment hires. In the later stages, unemployment hires become the dominant channel. This process captures the slow-moving reallocation forces that take time to materialize. As poaching becomes increasingly difficult toward the end, unemployment hires emerges as the primary source of diffusion. In the model, reallocation via unemployment more generally captures in a reduced form slow-moving reallocation channels. Appendix D.2 expands on this point.

Table 1: Steady-state properties in high and low mobility calibration.

	E-U rate	E-E rate	U-E rate	Initial s.s.	Final s.s.	Growth
High mobility (U.S.)	0.04	0.078	0.72	100%	123%	23.4%
Low mobility (EU)	0.02	0.04	0.38	102%	126%	23.4%

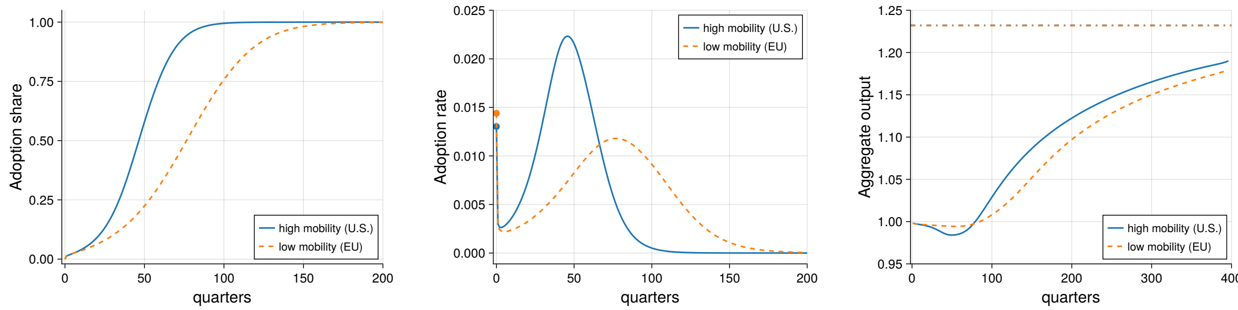
Notes: Initial steady state is relative to the high mobility (U.S.) economy. The growth rate is taken between initial and final steady-state. E-U denotes employment-to-unemployment rate, E-E employment-to-employment rate, and U-E unemployment-to-employment rate. Rates are quarterly.

4.3 Application: Divergence between U.S. and EU economies.

EU and U.S. growth rates have diverged substantially over the past two decades — by approximately one percentage point per year. Draghi (2024) provides a comprehensive investigation of the sources of this divergence. The analysis identifies a single source that primarily accounts for the trend: differential adoption and productivity growth in information and communication technologies (ICT). While the EU, and Germany in particular, maintain a productivity lead in mature industries characterized by incremental innovation, European countries lag behind in technology-intensive sectors exhibiting rapid technical change. This pattern is striking against the backdrop of well-documented differences in labor market institutions between these regions. We therefore investigate whether labor market institutions shape the diffusion of radical technologies and, through this channel, generate persistent cross-country differences in aggregate productivity and output.

I re-calibrate the model to lower mobility rates and longer job tenures as seen in the EU. Specifically, I re-calibrate the separation rate to a lower value and the matching efficiency while keeping the unemployment rate unchanged. This has the effect of lowering employment-to-employment transitions, while increasing job tenures, lowering unemployment-to-employment rates and increasing long-term unemployment. All of these are well-documented features in the EU labor markets (Borowczyk-Martins, 2025).

Steady-state. Lower mobility rates and longer job duration in the model are associated with modestly higher steady-state output as longer lasting matches facilitate on-the-job learning and the mutual transfer of knowledge. The model output in the initial and final steady states are given in Table 1. Total growth between steady-states is the same in the high and the low mobility calibration.



(a) Diffusion curve. **(b)** Adoption rate. **(c)** Aggregate output.

Figure 11: Adoption patterns for EU and U.S. calibrations.

Notes: High-mobility calibration corresponds to the U.S. calibration (blue, solid). Low-mobility & long tenure job calibration corresponds to the EU calibration (orange, dashed). The left panel plots the share of entrepreneurs that have adopted the new technology after its introduction. The middle panel plots the quarterly adoption rate. The right panel plots aggregate output over the transition, taking into account changes to knowledge variables, vacancy posting and fixed costs. Time $t = 0$ denotes the time the new technology vintage is introduced. Technological parameters are constant across countries. Only labor market parameters are re-calibrated.

Transition. While the high mobility (U.S.) and the low mobility (EU) calibrations are comparable in terms of steady-state output levels, they differ substantially in the diffusion of the new technology. Figure 11 plots the adoption and output paths for each calibration. In the low mobility, high tenure job (EU) calibration, diffusion is relatively slower for two reasons. First, solo entrepreneurs face a more binding constraint on skilled labor as access to workers with experience in the new vintage is more limited. Instead of poaching, solo entrepreneurs must wait for the labor to be reallocated via exogenous separations – a slower process. Moreover, due to long job tenures the reallocation channel through unemployment is slower also. Combined, these effects imply that the half-life is approximately 65% larger or 7-8 years longer in the low mobility (EU) than the high mobility (U.S.) calibration.

Figure 11c shows how differences in diffusion patterns translate to differential paths of aggregate output. While differences in technology intensity emerge relatively quickly, output paths take substantially longer to diverge. Further, absent changes in labor market institutions these differences are strongly persistent and once output gaps emerge catch-up is slow. This reveals that aggregate output can be an inadequate contemporaneous measure for the future economic performance early in the transition. In contrast, adoption rates are a more informative leading indicator.

Table 2: Net Present Value of Arrival of New Vintage

	high mobility (U.S.)	low mobility (EU)
Change in Welfare (% GDP)	26%	17.5%
Half life (quarters)	46.4	76.5

Welfare. How do the differential adoption rates and output paths under the respective labor market regimes map into welfare? To answer this question, I next compute the change in welfare attributable to the arrival of the new technology vintage. Since utility is transferable, the notion of changes in welfare corresponds to changes in net output relative to initial GDP. That is,

$$\Delta W_0 = \int_0^{+\infty} e^{-rt} \left(\frac{Y(t) - Y(0)}{Y(0)} \right) dt. \tag{9}$$

Numerical results are summarized in the Table 2 below. In total, the introduction of a new technology vintage increases the net present value of welfare in terms of initial steady-state GDP by 26% in the high mobility (U.S.) economy and only by 17.5% in the lower mobility (EU) economy. These effects are primarily driven by a longer half life of diffusion: reaching 50% penetration takes about 7 to 8 years longer in the low mobility economy.[36]

5 Policy and Efficiency

5.1 Social planner and inefficiencies.

I next turn to characterizing the social planner solution and the inefficiencies in the model, taking labor market frictions as given.[37] As a starting point, all decisions in the model are bilaterally efficient as a corollary of the sequential auction bargaining protocol. Therefore, all inefficiencies take the form of spillover externalities to third parties. There are two primary distortions that agents do not internalize. First, the

[36]The conclusions of this exercise draw parallels to Ljungqvist and Sargent (1998), who similarly find small differences in steady-state but very different (unemployment) dynamics between EU and U.S. economies. While their focus was on the welfare system, the present paper shares the point that the steady-state might conceal welfare losses from limited agility.

[37]Another friction in the model naturally stems from the search frictions in the labor market. As shown, these can have sizeable effects on welfare. In this section, we take labor market parameters as given.

search and matching framework gives rise to standard congestion externalities as in Hosios (1990). That is, entrepreneurs do not internalize that their vacancy posting increases the rate at which a single worker is contacted $p(\theta_t)$ and decreases the rate at which a single solo entrepreneur meets a worker $q(\theta_t)$. Likewise, workers do not internalize that their participation in search increases $q(\theta_t)$ while lowering the job $p(\theta_t)$. Second, agents do not internalize spillover gains of their actions to third parties. That is, when deciding whether to adopt a new technology vintage matches do not consider that this can facilitate adoption by solo entrepreneurs in the future. Appendix G illustrates these spillover externalities in an analytical model featuring three firms and a single worker.

To characterize the inefficiencies in the model I compute the set of optimal taxes and subsidies which implement the social planner allocation. This approach generalizes Fukui and Mukoyama (2025) to a setting with heterogeneous human capital, learning and technology choice. The proposition below characterizes the optimal set of transfers, followed a brief discussion of the underlying intuition

Proposition 1. *The decentralized equilibrium is generically inefficient. The optimal revenue-neutral transfers are*

$$\begin{aligned} T_t^e(z,h \mid \tau) &= \Gamma_t^e(z,h \mid \tau) - \frac{\xi}{\tilde{\mathbb{X}}_t}\mu_t \\ T_t^u(h \mid \tau) &= \Gamma_t^u(h \mid \tau) - \frac{1}{\tilde{\mathbb{X}}_t}\mu_t \\ T_t^v(z \mid \tau) &= -\frac{\tilde{v}_t(z \mid \tau)}{\tilde{\mathbb{V}}_t}\omega_t, \end{aligned}$$

where the positive spillover externalities are defined as

$$\begin{aligned} \Gamma_t^e(z,h \mid \tau) &= \xi\, p(\tilde{\theta}_t) \sum_{\tau' \in \mathcal{T}_t} \int \tilde{\Delta}_t^{EE}(z,h,z' \mid \tau,\tau')^{+} \frac{\tilde{v}_t(z' \mid \tau')}{\tilde{\mathbb{V}}_t} d\tilde{G}_t^v(z' \mid \tau') \\ \Gamma_t^u(h \mid \tau) &= p(\tilde{\theta}_t) \sum_{\tau' \in \mathcal{T}_t} \int \tilde{\Delta}_t^{UE}(h,z' \mid \tau,\tau')^{+} \frac{\tilde{v}_t(z' \mid \tau')}{\tilde{\mathbb{V}}_t} d\tilde{G}_t^v(z' \mid \tau'), \end{aligned}$$

and the congestion externality wedges are, letting $\eta(\theta) = p'(\theta)\theta/p(\theta)$ denote the elasticity of the aggregate *matching function with respect to the aggregate search input* $\mathbb{X}t$

$$\mu_t = \eta(\tilde{\theta}_t)\left(\sum_{\tau\in\mathcal{T}_t}\int \Gamma^e_t(z,h\mid\tau)d\tilde{G}^e_t(z,h\mid\tau) + \sum_{\tau\in\mathcal{T}_t}\int \Gamma^u_t(h\mid\tau)d\tilde{G}^u_t(h\mid\tau)\right)$$

$$\omega_t = \left(1-\eta(\tilde{\theta}_t)\right)\left(\sum_{\tau\in\mathcal{T}_t}\int \Gamma^e_t(z,h\mid\tau)d\tilde{G}^e_t(z,h\mid\tau) + \sum_{\tau\in\mathcal{T}_t}\int \Gamma^u_t(h\mid\tau)d\tilde{G}^u_t(h\mid\tau)\right).$$

Tilde variables represent the social planner counterparts to the competitive equilibrium variables.

Proposition 1 shows that vacancy posting is inefficiently high and is taxed at a strictly positive rate. This comes from the fact that solo entrepreneurs appropriate the entire surplus gain from poaching and hiring while imposing a congestion effect on other labor market participants. To correct this inefficiency, the social planner imposes a marginal tax on vacancy posting.

Matched entrepreneurs exert negative congestion externalities and positive spillover effects Γ^e_t via job-to-job mobility, which are not internalized in the present bargaining protocol and should be subsidized.[38] These spillover effects are increasing in labor market tightness θ_t, and depend on the expected surplus gains from hiring. Within a technology vintage, the spillover gains favor the formation of bridge jobs that help diffuse knowledge and reallocate to higher value jobs. For example, the social planner values more strongly the creation of high learning gain jobs as the human capital acquired on the job can later be used to be diffused to other entrepreneurs. A clean example is shown in the analytical model in G.

Across technology vintages, the spillover gains may favor the adoption of the new technology vintage if the worker is likely to subsequently facilitate the adoption by other entrepreneurs through job-to-job movements. These externalities are relatively larger early in the transition when the pool of previous adopters is small. Moreover, externalities are greater for technologies which feature a lower transferability of prior knowledge and thereby are subject to more severe constraints on adoption stemming from scarce skilled labor.

The presence of learning and technology diffusion breaks the tight connection between productivity

[38] This is in part an implication of the particular bargaining protocol. The generalization of the sequential auction bargaining protocol by Cahuc et al. (2006) allows matches to, at least partially, internalize these knowledge spillovers. Likewise, buyout offers are other institutional designs that allow some internalization of these externalities. However, these features would introduce other inefficiencies as solo entrepreneurs would be subject to the well-known hold-up problem.

and marginal value ladders as commonly used in job ladder models (Fukui & Mukoyama, 2025). As a result, it no longer necessarily holds that high productivity matches are overvalued in the competitive equilibrium relative to the social planner, as long as workers have the potential to diffuse knowledge elsewhere. In absence of knowledge spillovers, high productivity firms are located at the top of the value ladder. As workers at those firms are in the most valuable match, they do not move elsewhere and therefore spillover effects as small — while exerting the same congestion externalities on the labor market. In the presence of diffusive labor mobility, it could even be optimal to subsidize these high revenue firms, especially those experienced in the new technology vintage.

5.2 Subsidizing new technology adoption.

This section considers a particular subsidy to encourage the adoption of the new technology: a two-part policy (s, T). In this scheme, output from using the new technology vintage is subsidized at a rate s, but the scheme expires after T periods. The subsidy is financed by uniform revenue taxes and the budget is balanced in every period. That is, the after-tax flow revenue of an entrepreneur-worker match is

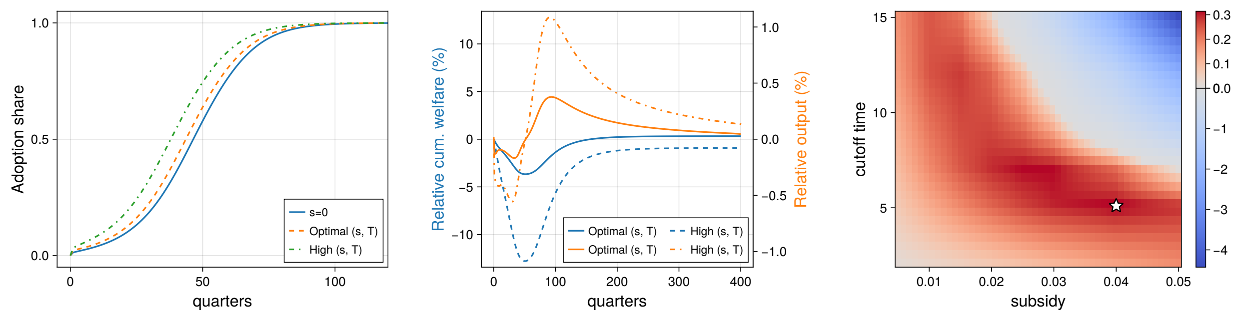
$$\mathcal{R}(z, h \mid \tau) = \begin{cases} (1 + s - tax_t)\pi(z, h \mid \tau) & \text{if } 0 \leq t \leq T \text{ and } \tau = 1, \\ (1 - tax_t)\pi(z, h \mid \tau) & \text{if } 0 \leq t \leq T \text{ and } \tau = 0, \\ \pi(z, h \mid \tau) & \text{if } t > T. \end{cases}$$

This subsidy generates incentives to adopt the new technology by increasing the expected revenues after adoption. Balance budget implies that in every period $0 \leq t \leq T$

$$tax_t = s \cdot \frac{\int \pi(z, h \mid 1) dG_t^e(z, h \mid 1) + \int \pi(z, 0 \mid 1) dG_t^v(z \mid 1)}{\sum_\tau \int \pi(z, h \mid \tau) dG_t^e(z, h \mid \tau) + \sum_\tau \int \pi(z, 0 \mid \tau) dG_t^v(z \mid \tau)}.$$

This formulation says that the effective subsidy received by adopters declines over the transition and expires at some time $t = T$. Therefore, the scheme is designed such that early adoption is incentivized and effective subsidies decay as $t \to \infty$.[39]

[39]Effective subsidy rates are the difference between $s - tax_t$.



(a) Diffusion curve. **(b)** Output and cumulative welfare. **(c)** Differential welfare heatmap.

Figure 12: Adoption subsidy on output with expiration time.

Notes: The left panel plots the adoption shares for three policies: (i) no policy baseline, i.e. $s = 0$ (ii) the optimal (s, T) menu and (iii) a high $(s, T) = (0.04, 12)$ schedule delivering faster adoption. The middle panel shows relative cumulative welfare (left axis) and relative output (right axis) over the transition. The right panel shows the differential net present value ΔW_0 for different policy menus (s, T). The white star denotes the optimal combination of (s^*, T^*).

Optimal subsidy & expiration-time policy. Figure 12 illustrates the effect of different policy schemes (s, T) on adoption patterns, aggregate output and cumulative welfare. The left panel demonstrates that positive subsidies accelerate technology diffusion by incentivizing early adoption. The middle panel highlights a critical intertemporal trade-off: accelerated adoption generates short-run output losses (orange, right axis) but yields long-run output gains, resulting in welfare losses over the initial quarters (blue, left axis). The magnitude of this trade-off depends on diffusion speed — overly rapid diffusion produces prohibitive short-run losses that future gains cannot offset, while moderate acceleration minimizes initial losses and generates positive, though modest, long-run welfare gains (solid, blue). The right panel maps the welfare differential ΔW_0 from technology introduction, as defined in (9), across the (s, T) policy space, with the optimal combination marked by the white star. This optimum targets early-adopting firms during the initial transition phase while incorporating a relatively short expiration horizon to balance the intertemporal welfare trade-offs.

Transitions can be too fast. These results highlight an important policy insight: accelerating diffusion is not unconditionally welfare-improving. In fact, the right panel of the above figure suggests that welfare losses from overly rapid diffusion, induced by large and persistent subsidy schemes, can be sizeable.

Intuitively, as subsidies expand, they induce adoption by progressively less productive matches. Since productivity in the new technology scales with existing capabilities, these marginal adopters lack sufficient profitability to justify the switch to the new technology.[40] From a welfare perspective, these firms would benefit more from delayed adoption, allowing them to first absorb knowledge spillovers from workers gaining experience at high-productivity early adopters. This learning externality creates a natural motive for measured, rather than maximal, diffusion speed.

Relation to inefficiencies. The preceding discussion on inefficiencies identified two main externalities in the model. First, matches fail to internalize the positive knowledge spillovers their workers generate when moving to future adopters when they decide whether to upgrade to the new technology. Second, firms do not fully internalize the negative congestion effects their labor market participation imposes. The strength of these externalities varies over the transition: knowledge spillovers are strongest early in the transition when the pool of non-adopters is largest, while congestion effects are strongest during the middle phase when competition for and poaching of diffusive workers intensifies. A time-limited subsidy targeted at pioneering adopters balances these intertemporal trade-offs.

6 Concluding remarks

This paper puts forth the idea that worker mobility is a central determinant of how path-breaking technologies diffuse through the economy, particularly during the early stages of the technology lifecycle. I show how skilled labor scarcity may create a binding constraint on technology adoption that endogenously generates the empirically-observed S-shaped diffusion curve and J-curve productivity patterns. Early in the transition, access to workers with new-vintage experience constrains adoption. As diffusion progresses and the pool of experienced workers expands, this constraint relaxes and adoption accelerates, progressively increasing the likelihood that non-adopters match with adopters. The analysis further highlights a fundamental trade-off facing policymakers: low mobility and high tenure job labor markets facilitate incremental learning in steady-state but impede the technological agility required during transformative

[40] Specifically, it does not justify them incurring the per-period fixed cost of operating the new technology while they are relatively more productive in the old technology.

transitions. Subsidies aimed at incentivizing adoption are optimally targeted at early adopters to set in motion the diffusion process and strike and intertemporal balance between competing externalities.

The framework developed in the present paper opens several avenues for future research. First, the current version assumes the exogenous arrival of a single new vintage. This could be relaxed along two dimensions in the pursuit of a unified account of growth: endogenizing the sequential development of these radical technologies and the co-existence of multiple, possibly competing, technologies. Second, the model could be extended to multi-worker firms to study the interaction between internal labor markets and organizational scale with external mobility in driving diffusion, while also capturing within-firm knowledge spillovers that may create heterogeneous technology intensity within firms. Third, the framework could be enriched to evaluate specific policy levers such as university education, dynamic entry and creative destruction as well as concrete training policies and international migration and visa programs. Finally, an important open question is the role of the bargaining protocol and surplus sharing between agents. Relaxing the assumption on bilateral efficiency could be an interesting avenue.

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Appendix

A Model appendix

A.1 Distributional dynamics

Denote lower case variables as the marginal distribution counterparts to their corresponding CDFs. The distribution over matches evolves in the interior of the state space[41]

$$
\begin{aligned}
\partial_t g_t^e(z,h \mid \tau) = &\underbrace{\mathcal{L}^*\left[g_t^e\right](z,h \mid \tau)}_{\text{Learning}} + \underbrace{\mathcal{Q}^*\left[g_t^e\right](z,h \mid \tau)}_{\text{Productivity shocks}} + \underbrace{\mathcal{M}\left[g_t^e\right](z,h \mid \tau; \boldsymbol{g}_t, t)}_{\text{Hiring \& poaching flows}} \\
&+ \underbrace{m_t^a(h)\delta(\tau-1)\delta(z-\underline{z}_t(h))}_{\text{Adoption}} - \underbrace{\kappa g_t^e(z,h \mid \tau)}_{\text{Separations}},
\end{aligned}
\tag{A.1}
$$

where $\boldsymbol{g}_t = \{g_t^e, g_t^v, g_t^u\}$ and $m_t^a(h)$ solves the flux at the boundary $(\underline{z}_t(h), h)$, i.e. the rate at which matched entrepreneurs hit the adoption threshold $\underline{z}_t(h)$[42]

$$
\begin{aligned}
m_t^a(h) = &\left(\alpha\left[h-\underline{z}_t(h)\right]^+ + \vartheta\left(\mu-\underline{z}_t(h)\right)\underline{z}_t(h)\right) g_t^e(\underline{z}_t(h), h \mid 0) - \frac{1}{2}\partial_z\left(\sigma^2 \underline{z}_t(h)^2 g_t^e(\underline{z}_t(h), h \mid 0)\right) \\
&- \underline{z}_t'(h)\alpha\left[\underline{z}_t(h)-h\right]^+ g_t^e(\underline{z}_t(h), h \mid 0).
\end{aligned}
\tag{A.2}
$$

The adjoint operators are given by

$$
\mathcal{L}^*\left[g_t^e\right](z,h \mid \tau) = -\partial_z\left(\alpha_1\left[h-z\right]^+ g_t^e(z,h \mid \tau)\right) - \partial_h\left(\left(\alpha_0 + \alpha_1\left[z-h\right]^+\right) g_t^e(z,h \mid \tau)\right) \tag{A.3}
$$

$$
\mathcal{Q}^*\left[g_t^e\right](z,h \mid \tau) = -\partial_z\left(\left(\vartheta(\mu-z)z\right) g_t^e(z,h \mid \tau)\right) + \frac{1}{2}\partial_{zz}\left(\sigma^2 z^2 g_t^e(z,h \mid \tau)\right) \tag{A.4}
$$

[41] That is for all $(z,h) \in \mathbb{R}^+ \times \mathbb{R}^+$ such that $z \geq \underline{z}_t(h)$. Note that this assumes that such a threshold exists. This is confirmed numerically in the calibrated version of the model.

[42] The probability flux through the boundary $\underline{z}_t(h)$ is given by

$$
R(h,t) = J(\underline{z}_t(h), h \mid 0) \cdot \mathbf{n}(\underline{z}_t(h), h) ds,
$$

where J denotes the probability flux vector at $(\underline{z}_t(h), h)$ from the KFE, $\mathbf{n} = \frac{(-\underline{z}_t'(h),1)}{\sqrt{1+\underline{z}_t'(h)}}$ the unit normal vector to the boundary at $(\underline{z}_t(h), h)$, and $ds = \sqrt{1+\underline{z}_t'(h)^2}dh$ the differential line element along the boundary.

$$\mathcal{M}^*\left[g_t^e\right](z,h\mid\tau;\mathbf{g}_t,t) = \underbrace{-\xi p(\theta_t)\sum_{\tau'\in\mathcal{T}_t}\int \mathbb{1}^{EE}(z,h,z'\mid\tau,\tau';t)\left(\frac{v_t(z'\mid\tau)}{\mathbb{V}_t}g_t^v(z'\mid\tau')\right)dz'g_t^e(z,h\mid\tau)}_{\text{Outflows due to E-E moves}}$$
$$+\underbrace{\xi p(\theta_t)\sum_{\tau_f,\tau_w\in\mathcal{T}_t}\iiint \mathcal{K}^{EE}((h',\tau_w,z',\tau_f),(h,\tau,z,\tau)\mid z'';t)g_t^e(z'',h'\mid\tau_w)dz''dh'\frac{v_t(z'\mid\tau_f)}{\mathbb{V}_t}g_t^v(z'\mid\tau_f)dz'}_{\text{Inflows due to E-E moves}}$$
$$+\underbrace{p(\theta_t)\sum_{\tau_f,\tau_w\in\mathcal{T}_t}\iiint \mathcal{K}^{UE}((h',\tau_w,z',\tau_f),(h,\tau,z,\tau);t)g_t^u(h'\mid\tau_w)dh'\frac{v_t(z'\mid\tau_f)}{\mathbb{V}_t}g_t^v(z'\mid\tau_f)dz'}_{\text{Inflows due to U-E moves}}. \tag{A.5}$$

The transition kernels $\mathcal{K}^{EE},\mathcal{K}^{UE}$ capture, conditional on meeting a worker, (1) whether a new match is formed (2) which technology vintage τ worker and firm agree to use and (3) the stochastic changes to the productivity vector due to the adoption decision. The kernel therefore captures the joint adoption and hiring decisions of worker-firm pairs as well as the stochastic changes to their productivity vector. The kernels are given by, letting δ denote the Dirac delta function,[43]

$$\mathcal{K}^{EE}=\begin{cases}\delta(h'-h)\delta(z'-z)\mathbb{1}\{\tau=0\} & \text{if } \mathbb{1}^{EE}(z'',h',z'\mid\tau_w,\tau_f;t)=1 \text{ and } \mathbb{1}^{A}(z',h'\mid\tau_f,\tau_w)=0\\ \mathcal{P}(\tau_w,\tau_f)\delta(h'-\overline{\delta}^{1-\tau_w}h)\delta(z'-\overline{\delta}^{1-\tau_f}z)\mathbb{1}\{\tau=1\} & \text{if } \mathbb{1}^{EE}(z'',h',z'\mid\tau_w,\tau_f;t)=1 \text{ and } \mathbb{1}^{A}(z',h'\mid\tau_f,\tau_w;t)=1\\ \left(1-\mathcal{P}(\tau_w,\tau_f)\right)\delta(h'-\underline{\delta}^{1-\tau_w}h)\delta(z'-\underline{\delta}^{1-\tau_f}z)\mathbb{1}\{\tau=1\} & \text{if } \mathbb{1}^{EE}(z'',h',z'\mid\tau_w,\tau_f;t)=1 \text{ and } \mathbb{1}^{A}(z',h'\mid\tau_f,\tau_w;t)=1\\ 0 & \text{if } \mathbb{1}^{EE}(z'',h',z'\mid\tau_w,\tau_f;t)=0,\end{cases} \tag{A.6}$$

and

$$\mathcal{K}^{UE}=\begin{cases}\delta(h'-h)\delta(z'-z)\mathbb{1}\{\tau=0\} & \text{if } \mathbb{1}^{UE}(h',z'\mid\tau_w,\tau_f;t)=1 \text{ and } \mathbb{1}^{A}(z',h'\mid\tau_f,\tau_w;t)=0\\ \mathcal{P}(\tau_w,\tau_f)\delta(h'-\overline{\delta}^{1-\tau_w}h)\delta(z'-\overline{\delta}^{1-\tau_f}z)\mathbb{1}\{\tau=1\} & \text{if } \mathbb{1}^{UE}(h',z'\mid\tau_w,\tau_f;t)=1 \text{ and } \mathbb{1}^{A}(z',h'\mid\tau_f,\tau_w;t)=1\\ \left(1-\mathcal{P}(\tau_w,\tau_f)\right)\delta(h'-\underline{\delta}^{1-\tau_w}h)\delta(z'-\underline{\delta}^{1-\tau_f}z)\mathbb{1}\{\tau=1\} & \text{if } \mathbb{1}^{UE}(h',z'\mid\tau_w,\tau_f;t)=1 \text{ and } \mathbb{1}^{A}(z',h'\mid\tau_f,\tau_w;t)=1\\ 0 & \text{if } \mathbb{1}^{UE}(h',z'\mid\tau_w,\tau_f;t)=0.\end{cases} \tag{A.7}$$

The distribution of solo entrepreneurs evolves as

$$\partial_t g_t^v(z\mid\tau) = \underbrace{\mathcal{Q}^*\left[g_t^v\right](z\mid\tau)}_{\text{Productivity shocks}} + \underbrace{\mathcal{H}^*\left[g_t^v\right](z\mid\tau;\boldsymbol{g}_t,t)}_{\text{Net hiring flows}} + \underbrace{\kappa\int g_t^e(z,h'\tau\mid\tau)dh'}_{\text{Inflows due to separations}}. \tag{A.8}$$

[43] The Dirac delta function is a generalized function such that $\int_{-\infty}^{+\infty}\delta(x)dx = 1$.

The adjoint operator for net hires $\mathcal{H}^*$ is

$$\begin{aligned}
\mathcal{H}^*\left[g_t^v\right](z \mid \tau; \boldsymbol{g}_t, t) = &- \underbrace{\xi p(\theta_t) \sum_{\tau \in \mathcal{T}_t} \iint \mathbb{1}^{EE}(z', h', z \mid \tau', \tau; t) g_t^e(z', h' \mid \tau') dh' dz' \left(\frac{v_t(z \mid \tau)}{\mathbb{V}_t} g_t^v(z \mid \tau) \right)}_{\text{Outflow due to successful E-E hires}} \\
&- \underbrace{p(\theta_t) \int \mathbb{1}^{UE}(h', z \mid \tau', \tau; t) g_t^u(h' \mid \tau') dh' \left(\frac{v_t(z \mid \tau)}{\mathbb{V}_t} g_t^v(z \mid \tau) \right)}_{\text{Outflows due to successful U-E hires}} \\
&+ \underbrace{\xi p(\theta_t) \sum_{\tau' \in \mathcal{T}_t} \int \mathbb{1}^{EE}(z, h, z' \mid \tau, \tau'; t) \left(\frac{v_t(z' \mid \tau')}{\mathbb{V}_t} g_t^v(z' \mid \tau') \right) dz' g_t^e(z, h \mid \tau) dh}_{\text{Inflows due to worker getting poached}} .
\end{aligned} \tag{A.9}$$

The evolution of the distribution of the unemployed is

$$\begin{aligned}
\partial_t g_t^u(h \mid \tau) = &\underbrace{-\alpha_u \partial_h g_t^u(h \mid \tau)}_{\text{Human capital depreciation}} + \underbrace{\kappa \int g_t^e(z', h \mid \tau) dz'}_{\text{Inflows due to separations}} \\
&\underbrace{-p(\theta_t) \sum_{\tau' \in \mathcal{T}_t} \int \mathbb{1}^{UE}(h, z' \mid \tau, \tau'; t) \frac{v_t(z' \mid \tau')}{\mathbb{V}_t} g_t^v(z' \mid \tau') dz' g_t^u(h \mid \tau)}_{\text{Outflows due to U-E move}} .
\end{aligned} \tag{A.10}$$

A.2 Poaching policies

By Equation (3.5) a worker is poached whenever $\mathbb{1}^{EE} = 1$. For illustrative purposes, let us consider the case without adoption. Then, poaching is successful whenever

$$S_t(z, h \mid 0) - S_t(z', h \mid 0) > V_t(z \mid 0) - V_t(z' \mid 0).$$

To understand the forces behind this equation let us consider each side of the equation in turn. The left-hand side satisfies

$$
\begin{aligned}
S_t(z,h \mid 0) - S_t(z',h \mid 0) = {} & \underbrace{\frac{\pi(z,h \mid 0) - \pi(z',h \mid 0)}{\rho+\kappa}}_{\text{static production}} \\
& + \underbrace{\frac{\mathcal{L}\left[S_t\right](z,h \mid 0) - \mathcal{L}\left[S_t\right](z',h \mid 0)}{\rho+\kappa}}_{\text{differential learning}} \\
& + \underbrace{\frac{\mathcal{Q}\left[S_t\right](z,h \mid 0) - \mathcal{Q}\left[S_t\right](z',h \mid 0)}{\rho+\kappa}}_{\text{differential exogenous shocks}} \\
& + \underbrace{\frac{\kappa}{\kappa+\rho}\big(V_t(z \mid 0) - V_t(z' \mid 0)\big)}_{\text{differential outside options}} \\
& + \underbrace{\frac{\partial_t S_t(z,h \mid 0) - \partial_t S_t(z',h \mid 0)}{\rho+\kappa}}_{\text{differential capital gains}}.
\end{aligned}
$$

The right-hand side satisfies

$$
\begin{aligned}
V_t(z \mid 0) - V_t(z' \mid 0) = {} & \underbrace{\frac{\pi(z,0 \mid 0) - \pi(z',0 \mid 0)}{\rho+d}}_{\text{differential production}} \\
& + \underbrace{\frac{\mathcal{Q}[V_t](z \mid 0) - \mathcal{Q}[V_t](z' \mid 0)}{\rho+d}}_{\text{differential exogenous shocks}} \\
& + \underbrace{\frac{\mathcal{H}(z \mid 0) - \mathcal{H}(z' \mid 0)}{\rho+d}}_{\text{differential hiring gains}} \\
& + \underbrace{\frac{\partial_t V_t(z \mid 0) - \partial_t V_t(z' \mid 0)}{\rho+d}}_{\text{differential capital gains}}.
\end{aligned}
$$

Noting that $\frac{\kappa}{\kappa+\rho} - 1 = -\frac{\rho}{\kappa+\rho}$, this delivers the following relation determining whether a worker is successfully poached

$$
\begin{aligned}
\left(\pi(z,h\mid 0)-\frac{\rho}{\rho+d}\pi(z,0\mid 0)\right)-\left(\pi(z',h\mid 0)-\frac{\rho}{\rho+d}\pi(z',0\mid 0)\right)& \\
+\mathcal{L}\left[S_t\right](z,h\mid 0)-\mathcal{L}\left[S_t\right](z',h\mid 0)& \\
+\mathcal{Q}\left[S_t-\frac{\rho}{\rho+d}V_t\right](z,h\mid 0)-\mathcal{Q}\left[S_t-\frac{\rho}{\rho+d}V_t\right](z',h\mid 0)& \\
+\frac{\rho}{\rho+d}\left(\mathcal{H}(z'\mid 0)-\mathcal{H}(z\mid 0)\right)& \\
+\partial_t\left(S_t(z,h\mid 0)-\frac{\rho}{\rho+d}V_t(z\mid 0)\right)-\partial_t\left(S_t(z',h\mid 0)-\frac{\rho}{\rho+d}V_t(z'\mid 0)\right)&>0.
\end{aligned}
$$

Interpretation. To understand these expressions it is instructive to consider limiting cases. First, suppose that $\zeta\to 1$ and $d\to 0$ such that $\pi(z,h\mid 0)=z+h$. In that case, there are no production complementarities and the expression in the first line is zero. That is, worker mobility is primarily pinned down by differential learning gains (second line) and differential abilities to hire a new worker (fourth line).[44] Another force operates through differential exposure to productivity shocks. As ζ falls production complementarities become the more dominant force while for $d>0$ more productive entrepreneurs have greater incentives avoiding the riskier outside option of being solo. That is, by becoming solo they would face the risk of exiting and receiving a value of zero, which is more costly to more productive entrepreneurs.

In another extreme, suppose that $z'=h$. In that case $\mathcal{L}\left[S_t\right](z',h\mid 0)=0$ and there are no learning gains available. We require $z\neq h$ for there to be positive learning gains.

Introducing adoption. The decision whether to poach or not now depends on — letting $\tau_f=0$ but $\tau_w=1$.

$$
\max\{S_t^A(z,h\mid 0,1)-S_t(z,h\mid 0),0\}+S_t(z,h\mid 0)-S_t(z,h\mid 1)>V_t(z\mid 1)-V_t(z\mid 1).
$$

[44] This in turn is pinned down by the offer distribution of marginal values to the entrepreneur, i.e. the distribution of positive Δ^{EE} and Δ^{UE}.

In total

$$
\begin{aligned}
&(\rho+\kappa)\max\{S_t^A(z,h\mid 0,1)-S_t(z,h\mid 0),0\}\\
&\quad+\left(\pi(z,h\mid 0)-\frac{\rho}{\rho+d}\pi(z,0\mid 0)\right)-\left(\pi(z',h\mid 1)-\frac{\rho}{\rho+d}\pi(z',0\mid 1)\right)\\
&\quad+\mathcal{L}\left[S_t\right](z,h\mid 0)-\mathcal{L}\left[S_t\right](z',h\mid 1)\\
&\quad+\mathcal{Q}\left[S_t-\frac{\rho}{\rho+d}V_t\right](z,h\mid 0)-\mathcal{Q}\left[S_t-\frac{\rho}{\rho+d}V_t\right](z',h\mid 1)\\
&\quad+\frac{\rho}{\rho+d}\left(\mathcal{H}(z'\mid 1)-\mathcal{H}(z\mid 0)\right)\\
&\quad+\partial_t\left(S_t(z,h\mid 0)-\frac{\rho}{\rho+d}V_t(z\mid 0)\right)-\partial_t\left(S_t(z',h\mid 1)-\frac{\rho}{\rho+d}V_t(z'\mid 1)\right)>0.
\end{aligned}
$$

There are two opposing forces: (1) there are additional gains from poaching due to the option value to adopt, captured by $(\rho+\kappa)\max\{S_t^A(z,h\mid 0,1)-S_t(z,h\mid 0),0\}$, but (2) it becomes relatively harder to poach since the marginal value of the worker to a matched entrepreneur now is scaled. This is seen in a straightforward manner for the limit case in which $\zeta=1$ and $\pi(z,h\mid 1)-\pi(z,0\mid 1)=A(z+h)-k_f-(Az-k_f)=Ah$, which scales in A. Further, in the limit as $\xi\to 1$ we get that

$$
\pi(z,h\mid 0)-\frac{\rho}{\rho+d}\pi(z,0\mid 0)-\left(\pi(z',h\mid 1)-\frac{\rho}{\rho+d}\pi(z',0\mid 1)\right)=-(A-1)h-\frac{d}{\rho+d}\left(Az'-k_f-z\right).
$$

A.3 Hiring policies

By Equation (3.5) a worker is poached whenever $\mathbb{1}^{UE}=1$. For illustrative purposes let us consider the case without adoption. Then, hiring is successful if and only if

$$
S_t(z,h\mid 0)-V_t(z\mid 0)-U_t(h\mid 0)>0.
$$

That is, if the match generates a positive surplus.

The left-hand side is given by

$$
\begin{aligned}
(\rho+\kappa)\left(S_t(z,h\mid 0)-V_t(z\mid 0)-U_t(z\mid 0)\right)= &\underbrace{\pi(z,h\mid 0)-\frac{\rho}{\rho+d}\pi(z,0\mid 0)-bh}_{\text{static surplus}}\\
&+\underbrace{\mathcal{L}\left[S_t\right](z,h\mid 0)}_{\text{Learning}}\\
&-\underbrace{\left(-\frac{\alpha_u}{\rho+\alpha_u}bh\right)}_{\text{Human capital depreciation}}\\
&+\underbrace{\mathcal{Q}\left[S_t\right](z,h\mid 0)-\frac{\rho}{\rho+d}\mathcal{Q}\left[V_t\right](z\mid 0)}_{\text{exogenous shocks}}\\
&-\underbrace{\frac{\rho}{\rho+d}\mathcal{H}(z\mid 0)}_{\text{option value of hiring}}\\
&+\underbrace{\partial_t\left(S_t(z,h\mid 0)-\frac{\rho}{\rho+d}V_t(z\mid 0)\right)}_{\text{capital gains}}\\
\geq &0.
\end{aligned}
$$

In the limit of no production complementarities the first line becomes $(1-b)h$, which is positive as long as $b<1$. The second line denotes the learning gains. The third line denotes the cost of unemployment through human capital depreciation. The fourth line captures the exogenous shocks. The forth line denotes the opportunity cost as the solo entrepreneur foregoes the option value of hiring a different worker.

A.4 Adoption decision

A.4.1 Internal adoption

For simplicity let $\delta = \bar{\delta} = \underline{\delta}$. Then,

$$
\begin{aligned}
S_t(\delta z, \delta h \mid 1) - S_t(z, h \mid 0) = & \underbrace{\frac{\pi(\delta z, \delta h \mid 1) - \pi(z, h \mid 0)}{\rho + \kappa}}_{\text{Short-run output differences}} \\
& + \underbrace{\frac{\mathcal{L}\left[S_t\right](\delta z, \delta h \mid 1) - \mathcal{L}\left[S_t\right](z, h \mid 0)}{\rho + \kappa}}_{\text{Differential learning gains}} \\
& + \underbrace{\frac{\mathcal{Q}\left[S_t\right](\delta z, \delta h \mid 1) - \mathcal{Q}\left[S_t\right](z, h \mid 0)}{\rho + \kappa}}_{\text{Differential productivity shocks}} \\
& + \underbrace{\frac{\kappa}{\rho + \kappa}\left(V_t(\delta z \mid 1) - V_t(z \mid 0)\right)}_{\text{Differential changes in outside value for entrepreneur}} \\
& + \underbrace{\frac{\kappa}{\rho + \kappa}\left(U_t(\delta h \mid 1) - U_t(h \mid 0)\right)}_{\text{Differential changes in outside value for entrepreneur}} .
\end{aligned}
\tag{A.11}
$$

Note that $\pi(\delta z, \delta h \mid 1) - \pi(z, h \mid 0) = (A\delta - 1)\,\pi(z, h \mid 0) - k_f$. If this is positive, then adoption is immediately profitable. If not, then adoption may still occur through the joint learning and productivity shock process. First, mean reversion implies a form of exogenous learning whenever $\log z$ falls below the long-run mean. Second, the volatility harvesting process implies that match duration induces a positive trend in z and h jointly. This may make adoption attractive even if it delivers short-run losses. Third, if an entrepreneur adopts with a highly-skilled worker, i.e. large h, then this learning gain persists in the new technology and increase the trend growth of the entrepreneur's productivity.

The fourth line captures the trade-off between (1) insurance from wide-spread adoption and (2) free-riding mechanism. When the pool of adopters is large then it is easier to find a previous adopter when solo and can immediately form more productive matches relative to the counterfactual of a small pool of such workers. That is, these forces increase $V_t(\delta z \mid 1)$. On the other hand, a large pool of such workers also increases the opportunity of adoption via hiring which is less costly to the entrepreneur. These forces are traded off as $V_t(\delta z \mid 1) - V_t(z \mid 0)$ and discounted by the likelihood of an exogenous separation.

Further, the only way aggregate conditions matter here is through the outside option V_t. The presence of long-lasting vacancies, i.e. vacancy chains, is how the result in Lise and Robin (2017) that surplus is independent of aggregate distributions is broken.

A.4.2 Hire to adopt

Let us now consider when a new match with initial states (z, τ_f) and (h, τ_w) decides to use the new technology when $\tau_f = 0$ and $\tau_w = 1$. Then,

$$
\begin{aligned}
S_t(\delta z, h \mid 1) - S_t(z, h \mid 0) = & \underbrace{\frac{\pi(\delta z, h \mid 1) - \pi(z, h \mid 0)}{\rho + \kappa}}_{\text{Short-run output differences}} \\
& + \underbrace{\frac{\mathcal{L}\left[S_t\right](\delta z, h \mid 1) - \mathcal{L}\left[S_t\right](z, h \mid 0)}{\rho + \kappa}}_{\text{Differential learning gains}} \\
& + \underbrace{\frac{\mathcal{Q}\left[S_t\right](\delta z, h \mid 1) - \mathcal{Q}\left[S_t\right](z, h \mid 0)}{\rho + \kappa}}_{\text{Differential productivity shocks}} \\
& \underbrace{+ \frac{\kappa}{\rho + \kappa}\left(V_t(\delta z \mid 1) - V_t(z \mid 0)\right)}_{\text{Differential changes in outside value for entrepreneur}} \quad .
\end{aligned}
\tag{A.12}
$$

The major difference relative to above is that since the worker has already accumulated technology-specific expertise only the entrepreneur is subject to the limited transferability. As a result, learning gains open up which further incentivize the adoption of the technology — as captured by the second line.

A.5 Decomposition of new adopters

A.5.1 Newly adopting entrepreneurs

Let the mass of entrepreneurs using the new technology be denoted by μ_t^E and the share by μ_t^E/M. Then,

$$
\begin{aligned}
\dot{\mu}_t^E =& \underbrace{\int m_t^a(h)\delta(z - \underline{z}_t(h))dhdz}_{\text{internal adoption}} \\
&+ \underbrace{\xi p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{v_t(z \mid 0)}{\mathbb{V}_t} \int \mathbb{1}^{EE}\left(z', h', z \mid \tau, 0; t\right) \mathbb{1}^{A}\left(h', z \mid \tau, 0; t\right) dG_t^e(z', h' \mid \tau) dG_t^v(z \mid 0)}_{\text{Poach to adopt}} \\
&+ \underbrace{p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{v_t(z \mid 0)}{\mathbb{V}_t} \int \mathbb{1}^{UE}\left(h', z \mid \tau, 0; t\right) \mathbb{1}^{A}\left(h', z \mid \tau, 0; t\right) dG_t^u(h' \mid \tau) dG_t^v(z \mid 0)}_{\text{Hire unemp. to adopt}} \\
&- \underbrace{\xi p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{v_t(z \mid 0)}{\mathbb{V}_t} \int \mathbb{1}^{EE}\left(z', h', z \mid \tau, 0; t\right) \left(1 - \mathbb{1}^{A}\left(h', z \mid \tau, 0; t\right)\right) dG_t^e(z', h' \mid \tau) dG_t^v(z \mid 1)}_{\text{Downgrade technology}} \\
&- \underbrace{p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{v_t(z \mid 0)}{\mathbb{V}_t} \int \mathbb{1}^{UE}\left(h', z \mid \tau, 0; t\right) \left(1 - \mathbb{1}^{A}\left(h', z \mid \tau, 0; t\right)\right) dG_t^u(h' \mid \tau) dG_t^v(z \mid 1)}_{\text{Downgrade technology}}
\end{aligned}
\tag{A.13}
$$

A.5.2 Newly adopting workers

Let the mass of workers using the new technology be denoted by μ_t^W which is equivalent to its share since the population of workers is normalized to one. Then,

$$
\begin{aligned}
\dot{\mu}_t^W = & \underbrace{\int m_t^a(h)\delta(z - \underline{z}_t(h))dhdz}_{\text{internal adoption}} \\
& \underbrace{+ \xi p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{\upsilon_t(z \mid \tau)}{\mathbb{V}_t} \int \mathbb{1}^{EE}\left(z', h', z \mid 0, \tau; t\right) \mathbb{1}^{A}\left(h', z \mid 0, \tau; t\right) dG_t^e(z', h' \mid 0) dG_t^\upsilon(z \mid \tau)}_{\text{Poached \& adopt}} \\
& \underbrace{+ p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{\upsilon_t(z \mid \tau)}{\mathbb{V}_t} \int \mathbb{1}^{UE}\left(h', z \mid 0, \tau; t\right) \mathbb{1}^{A}\left(h', z \mid 0, \tau; t\right) dG_t^u(h' \mid \tau) dG_t^\upsilon(z \mid 0)}_{\text{Hired \& adopt}} \\
& \underbrace{- \xi p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{\upsilon_t(z \mid \tau)}{\mathbb{V}_t} \int \mathbb{1}^{EE}\left(z', h', z \mid 1, \tau; t\right) \left(1 - \mathbb{1}^{A}\left(h', z \mid 1, \tau; t\right)\right) dG_t^e(z', h' \mid 1) dG_t^\upsilon(z \mid \tau)}_{\text{Downgrade technology}} \\
& \underbrace{- \xi p(\theta_t) \sum_{\tau\in\{0,1\}} \int \frac{\upsilon_t(z \mid \tau)}{\mathbb{V}_t} \int \mathbb{1}^{UE}\left(h', z \mid 1, \tau; t\right) \left(1 - \mathbb{1}^{A}\left(h', z \mid 1, \tau; t\right)\right) dG_t^u(h' \mid 1) dG_t^\upsilon(z \mid \tau)}_{\text{Downgrade technology}}
\end{aligned}
\tag{A.14}
$$

A.6 Further details on learning dynamics

In this section, I show that for matches that persist forever, h and z grow unboundedly. Growth comes from systematically capturing upside volatility. A heuristic argument can be made as follows. Solving the SDE

$$
\mathbb{E}_t\left[\log h(t)\right] = \log h(0) + \alpha \int_0^t \left[z(s)/h(s) - 1\right]^+ ds. \tag{A.15}
$$

Since $dz(t)$ is an Ornstein-Uhlenbeck process, for any scalar $L \in \mathbb{R}$ $z(s)$ almost surely crosses L infinitely times. Hence, for any $L \geq h(s)$ there exists some T such that $z(T) > h(s)$. The worker learns from the firm and h grows. Since z moves continuously these crossings contribute positive areas and these non-zero contributions make $h(t) \to \infty$. Then, since $h(t)$ grows unboundedly the following will as well since

$$
\mathbb{E}_t\left[\log z(t)\right] = e^{-\vartheta t} \log z(0) + \alpha \int_0^t e^{-\vartheta(t-s)} \left[h(s)/z(s) - 1\right]^+ ds. \tag{A.16}
$$

A.7 Special case: Long-run

Let the vacancy posting cost and the unemployment benefit b scale in aggregate output Y_t. Let $d = 0$. Then,

$$\lim_{t\to\infty} S_t(z, h \mid 1) = AS_{ss}(z, h \mid 0) - k_f/\rho \tag{A.17}$$

and

$$\lim_{t\to\infty} V_t(z, h \mid 1) = AV_{ss}(z, h \mid 0) - k_f/\rho \tag{A.18}$$

and

$$\lim_{t\to\infty} U_t(h \mid 1) = AU_{ss}(h \mid 0) \tag{A.19}$$

All decisions are unchanged, and therefore the same distribution exists. So can chain these together giving long-run growth through these waves.

B Calibration Appendix

We map our empirical estimate to the average probabilty of an employed worker moving to a newly adoption firm in period t. This is informative of degree to which poaching is related to adoption. Formally, it is given by

$$\int \left[\xi p(\theta_t) \left(\frac{v_t(z \mid 0)}{\mathbb{V}_t} \right) \int \mathbb{1}^{EE}(z', h, z \mid 1, 0) \mathbb{1}^{A}(z, h \mid 0, 1) \frac{dG^e_t(z', h \mid 1)}{\int dG^e_t(z', h \mid 1)} \right] \frac{dG^v_t(z \mid 0)}{\int dG^v_t(z \mid 0)}.$$

We show the time path of this transition in Figure B.1, which is similar to our empirical estimates for senior workers in the early stages of the transition.

The full calibration table is shown below

Table B.1: Calibrated parameters

Parameter	Description	Value	Comment
		General parameters	
ρ	discount rate	0.02	8% annual
M	mass of entrepreneurs	1.2	-
d	exit rate (solo entrepreneurs)	κ	-
		Labor market parameters	
χ	matching efficiency	0.725	unemployment rate of 5%
β	matching elasticity	0.5	standard
ξ	relative search intensity (J2J)	0.215	EE rate
κ	exogenous separation rate	0.038	EU rate
b	benefits	0	-
		Entrepreneur's parameters	
ζ	production complementarities	0.8	Herkenhoff et al. (2024)
σ^2	volatility	0.005	90-10 ratio
ϑ	persistence	0.02	-
μ	long-run mean	$e^{-\sigma^2/(2\vartheta)}$	normalization
$\bar{c}$	vacancy cost scaling	1	normalization
γ_v	convexity of vacancy cost	3.450	Bilal et al. (2022)
		Learning parameters	
α	learning rate	0.01	Herkenhoff et al. (2024)
α_u	human capital depreciation	0.032	Herkenhoff et al. (2024)
		Transition parameters	
A	lower mc	1.5	exercise
k_f	higher fixed cost	0.225	-
$\overline{\delta}$	high transferability	0.915	-
$\underline{\delta}$	low transferability	0.715	-
$\mathcal{P}(0,0)$		0	normalization
$\mathcal{P}(1,0)$		0.4	empirical target
$\mathcal{P}(0,1)$		0.4	symmetry

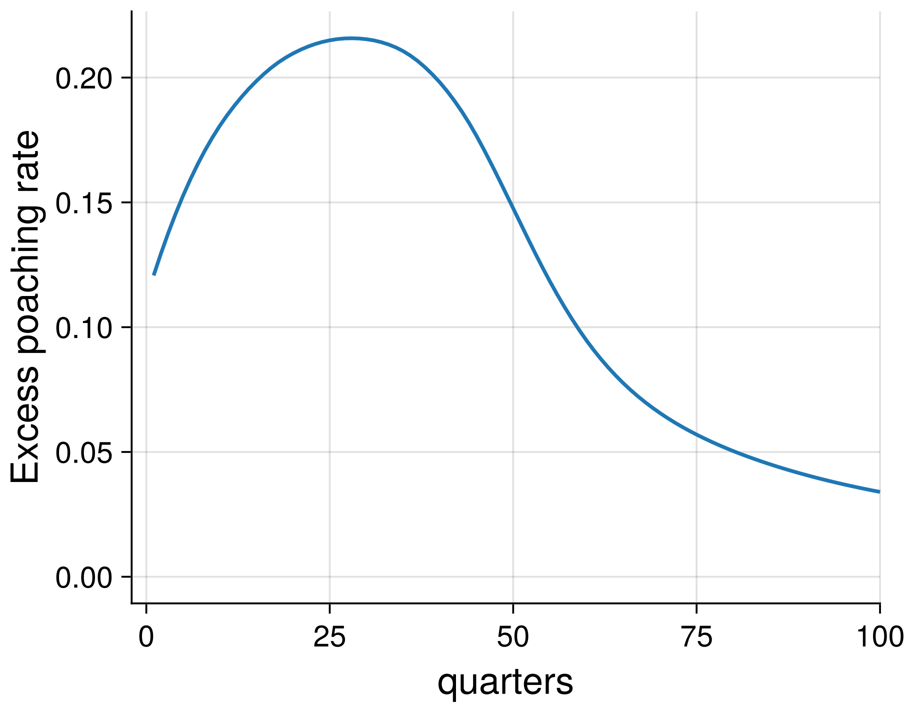


Figure B.1: Probability of moving to non-adopter.

C Empirical Appendix

C.1 Distribution of adoption years

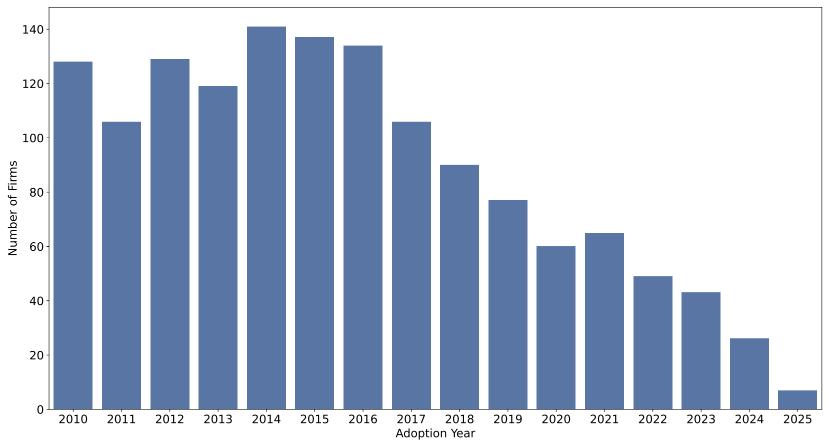


Figure C.2: Distribution of adoption years of Compustat sample (2010-2024)

C.2 Share of external AI hires

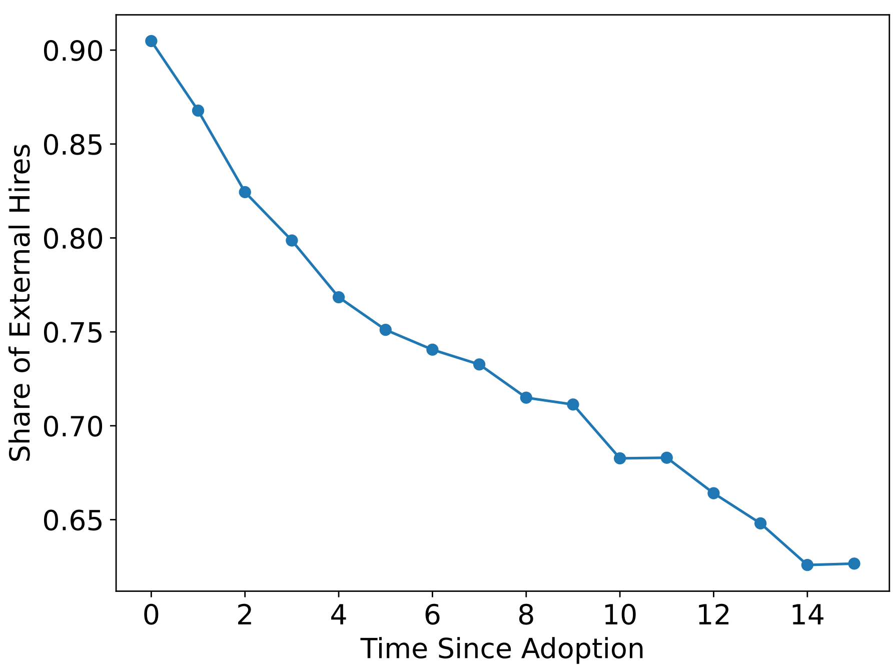


Figure C.3: Share of external hires of AI jobs

C.3 Subsample restrictions.

The following subsample restrictions are applied to the raw data. The main analysis covers years 2010 to 2024, a time period with good coverage and substantial AI adoption. The distribution of AI adoption during this sample is shown in Figure C.2. I restrict the focus to professional workers (SOCs 11-19) in metropolitan MSAs working in professional services, tech, and finance industries and high-tech manufacturing.[45] The final analysis uses worker flows where the destination firm corresponds to a publicly traded firm from Compustat while the origin firm is public or private. This allows us to retain as much information as possible regarding the movement of workers while supplementing the data with additional information on e.g. firm sales.

C.4 Identification argument

How should we think about the correlations documented in Section 2? We will now discuss the identification argument through which we can view the presented correlation. A classic concern in empirical studies of technology adoption is that selection into adoption is non-random and endogenous to the expected return of using the technology. To address this issue, I follow an approach similar to Humlum (2019). Specifically, the regression in (2) includes as controls pre-adoption sales per worker. The crucial assumption is that revenue productivity (growth) and industry-firm fixed effects are good proxies for a firm's expected return of adoption. Then, conditional on those controls, access to highly skilled workers with prior AI experience is a binding constraint for adoption and firms successfully implement AI only if they match with the right type of worker – a process that is at least in part random and an integral feature of canonical labor search models. I use this interpretation to map our empirical estimates to the quantitative model.

C.5 Robustness

The estimates from Section 2 are robust to a series of variations.

C.5.1 Excluding initial AI adopters

A concern is that we use labor market statistics to both identify adoption events and measure worker flows. To mitigate the extent to which this relationship could be mechanical in nature, I next exclude the workers which are used to identify adoption events from the analysis. That is, all AI workers that appear in a firm during the first year that AI is used in the firm are excluded from the analysis for that year. Estimates are largely unchanged.

[45] The included industry codes (NAICS) are 51, 52, 54, 55, 3341, 3342, 3344, 3345, 3364.

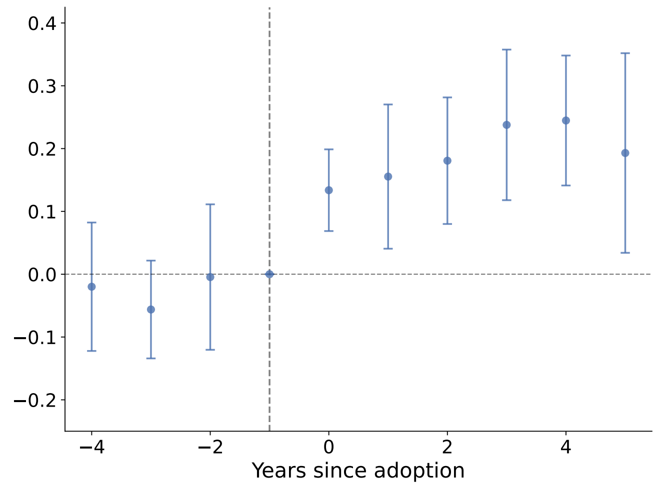


Figure C.4: Excess probabilities. Excluding first AI workers in firm.

Notes: This graph plots the coefficient estimates β_h from (2) and corresponds to how excess probabilities $\gamma_{cj\ell t+h}$ change with distance to the adoption year t. This version excludes first AI adopters in a firm. Estimated on subsample of E-E movers at annual frequency. The regression controls for industry-year and location fixed effects as well as pre-adoption variables related to selection into adoption (e.g. sales per worker). The sample period is 2010-2024. Destination firms in the sample are Compustat firms. Origin firms in the sample is the universe of firms in Revelio Labs. Standard errors are clustered at the firm-year level.

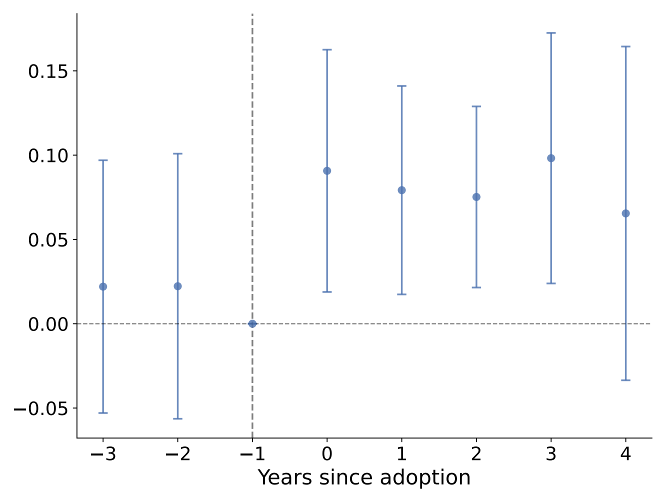
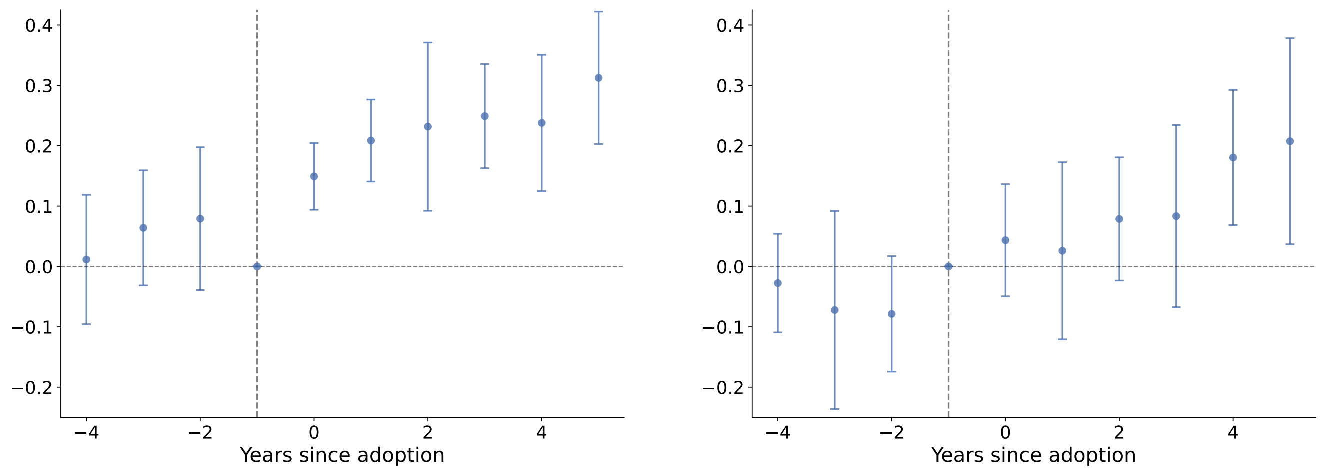


Figure C.5: Excess probabilities. Including firm-level fixed effects. Excluding first AI workers in firm.

Notes: This graph plots the coefficient estimates β_h from (2) and corresponds to how excess probabilities $\gamma_{cj\ell t+h}$ change with distance to the adoption year t. This version excludes first AI adopters in a firm. Estimated on subsample of E-E movers at annual frequency. The regression controls for firm-level, industry-year and location fixed effects as well as pre-adoption variables related to selection into adoption (e.g. sales per worker). The sample period is 2010-2024. Destination firms in the sample are Compustat firms. Origin firms in the sample is the universe of firms in Revelio Labs. Standard errors are clustered at the firm-year level.



(a) Senior

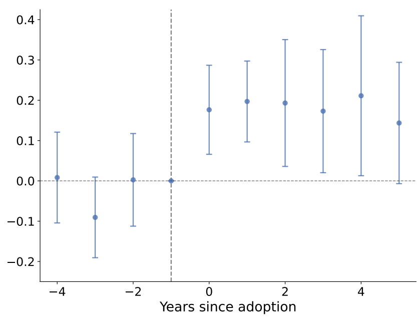
(b) Junior

Figure C.6: Excess probabilities by seniority status. Excluding first AI workers in firm.

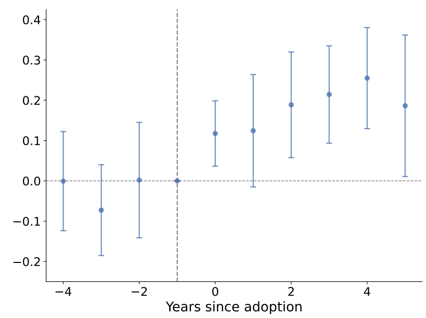
Notes: This graph plots the coefficient estimates β_h^c from (2) and corresponds to how excess probabilities $\gamma_{cj\ell t+h}$ change with distance to the adoption year t, separately for two subgroups c: senior workers (e.g. managers and directors) and junior workers (e.g. analysts). Excluding first AI adopters in firm. The left panel shows coefficients for the senior subsample. The right panel shows coefficients for the junior subsample. Estimated on subsample of E-E movers at annual frequency. The regression controls for industry-year and location fixed effects as well as pre-adoption variables related to selection into adoption (e.g. sales per worker). The sample period is 2010-2024. Destination firms in the sample are Compustat firms. Origin firms in the sample is the universe of firms in Revelio Labs. Standard errors are clustered at the firm-year level.

C.5.2 Other event studies

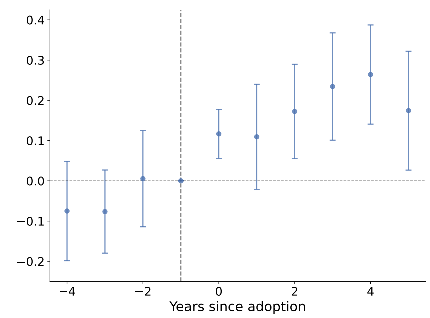
We explore two other specifications of the baseline regression in Section 2.3. In Figure C.7d we show the estimated coefficients of (2) where we additionally include destination firm fixed effects α_j. In Figure C.7e we show the estimated coefficients of a modified version of (1) where c denotes the set of workers from all firms that have previously utilized AI. The indicator D_{ikt} now takes the value one if the worker has previously held an AI position. The estimates indicate that it is really the AI adopters that see the increase in the hiring.



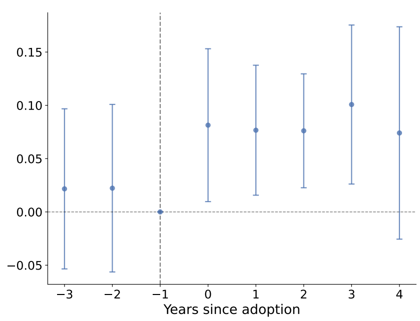
(a) Focussed occupations.



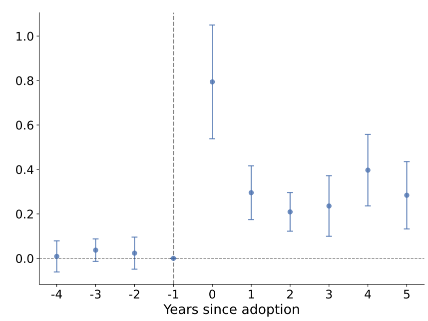
(b) Focussed MSAs.



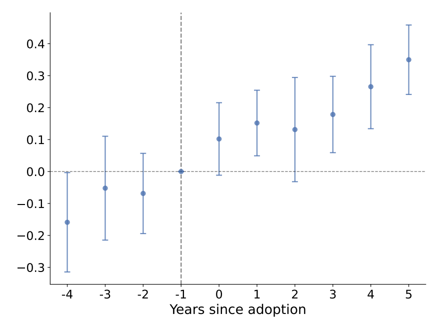
(c) Excluding IT sector.



(d) Including firm fixed effects.



(e) AI workers from previous adopters.



(f) Hiring from other industries.

Figure C.7: Alternative specification robustness.

D Additional model results

D.1 Free-riding motives over transition

TO BE COMPLETED.

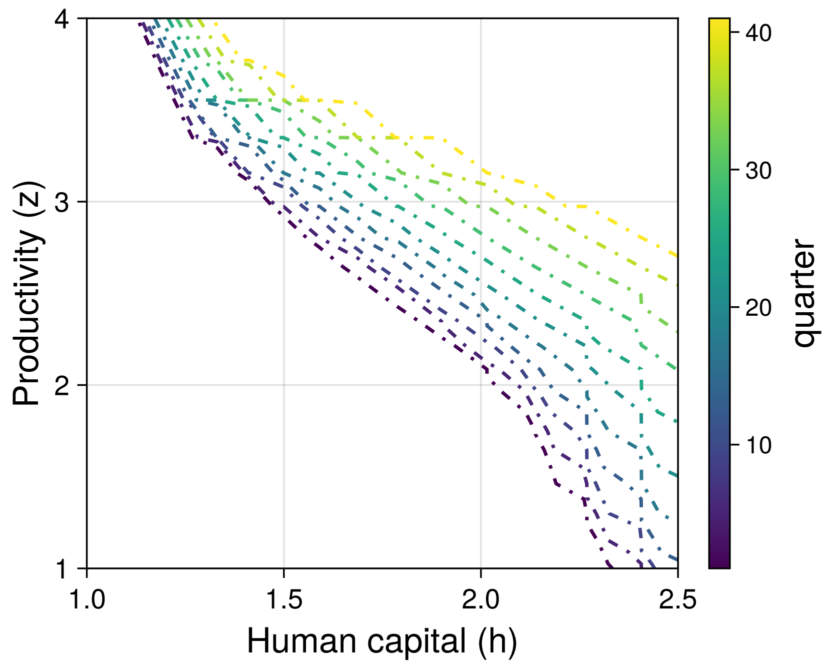


Figure D.8: Adoption thresholds over the transition.

D.2 Increased skill supply via education

Consider the following extension. Workers retire at rate ϱ at which point they are replaced by new workers who enter with the same level of human capital h, but may be biased towards the new technology vintage $\tau = 1$. Specifically, a new worker replacing a retiree with human capital h enters as trained in the new technology with density $f_t(h \mid 1)$, where

$$f_t(h \mid 1) = \min\left\{\omega, \frac{1-u_t}{\iint g_t^e(z,h \mid 1)}\right\} \int g_t^e(z,h \mid 1)dz + g_t^u(h \mid 1).$$

The min operator ensures that densities will be positive for $f_t(h \mid 0)$, but will only be binding towards the end of the transition or for very large values of ω. Let the density for entering with the old technology be given by

$$f_t(h \mid 0) = \frac{\max\left\{1-u_t-\omega \iint g_t^e(z,h \mid 1)dzdh, 0\right\}}{\iint g_t^e(z,h \mid 0)dzdh} \int g_t^e(z,h \mid 0)dz + g_t^u(h \mid 0).$$

It can be verified that indeed $\int f_t(h \mid 0)dh + \int f_t(h \mid 1)dh = 1$, with $f_t(h \mid \tau) \geq 0$ for all $h, \tau \in \mathbb{R}^+ \times \{0,1\}$ and this is a valid density.

Comments.. The above representation captures in a reduced form endogenous education decisions by workers. If there are more employment opportunities in the new technology for a specific worker h, i.e. a larger $\int g^e_t(z, h \mid 1)dz$, they are more likely to acquire these skills during the educational years.[46] Alternatively, we can simply view the parameter ω as an exogenous force increasing the labor supply of new workers entering the labor market.

Distributional dynamics. Adapting the distributional dynamics form (A.10) to the current setting gives

$$\begin{aligned}
\partial_t g^u_t(h \mid \tau) = \; & \underbrace{-\alpha_u \partial_h g^u_t(h \mid \tau)}_{\text{Human capital depreciation}} \; \underbrace{+\kappa \int g^e_t(z', h \mid \tau)dz'}_{\text{Inflows due to separations}} \\
& \underbrace{-p(\theta_t) \sum_{\tau' \in \mathcal{T}_t} \int \mathbb{1}^{UE}(h, z' \mid \tau, \tau'; t) \frac{v_t(z' \mid \tau')}{\mathbb{V}_t} g^v_t(z' \mid \tau')dz' g^u_t(h \mid \tau)}_{\text{Outflows due to U-E move}} \\
& \underbrace{-\varrho g^u_t(h \mid \tau) + \varrho f_t(h \mid \tau)}_{\text{net retirement flows}}.
\end{aligned} \tag{D.20}$$

Specifically, for $\tau = 1$ this reduces to

$$\begin{aligned}
\partial_t g^u_t(h \mid 1) = \; & \underbrace{-\alpha_u \partial_h g^u_t(h \mid 1)}_{\text{Human capital depreciation}} \\
& \underbrace{+\left(\kappa + \varrho \min\left\{\omega, \frac{1-u_t}{\iint g^e_t(z, h \mid 1)}\right\}\right) \int g^e_t(z', h \mid 1)dz'}_{\text{Inflows due to separations}} \\
& \underbrace{-p(\theta_t) \sum_{\tau' \in \mathcal{T}_t} \int \mathbb{1}^{UE}(h, z' \mid 1, \tau'; t) \frac{v_t(z' \mid \tau')}{\mathbb{V}_t} g^v_t(z' \mid \tau')dz' g^u_t(h \mid 1)}_{\text{Outflows due to U-E move}}.
\end{aligned} \tag{D.21}$$

[46]For expositional simplicity, the endogenous education choice is not modelled explicitly here.

For $\tau = 0$ we get

$$\begin{aligned}\partial_t g^u_t(h \mid 0) = & \underbrace{-\alpha_u \partial_h g^u_t(h \mid 0)}_{\text{Human capital depreciation}} \\ & + \underbrace{\left(\kappa + \varrho \frac{\max\left\{1 - u_t - \omega \iint g^e_t(z,h \mid 1) dz dh, 0\right\}}{\iint g^e_t(z,h \mid 0) dz dh}\right) \int g^e_t(z',h \mid 0) dz'}_{\text{Inflows due to separations}} \\ & \underbrace{-p(\theta_t) \sum_{\tau' \in \mathcal{T}_t} \int \mathbb{1}^{UE}(h, z' \mid 0, \tau'; t) \frac{v_t(z' \mid \tau')}{\mathbb{V}_t} g^v_t(z' \mid \tau') dz' g^u_t(h \mid 0)}_{\text{Outflows due to U-E move}} .\end{aligned} \tag{D.22}$$

From the above expressions, it becomes clear that when looking at unemployment dynamics an exogenous increase in the skill supply of new workers is isomorphic to separation rates which depend on the technology vintage. In the special case where $\omega = 1$, the distributional dynamics above would be isomorphic to the version presented in the main text. Away from the special case, i.e. $\omega > 1$, the above expression acts to increase the overall supply of skilled workers but does so only slowly as the stock of workers experienced in the new technology increases.

Taking stock. Unemployment in the model in the main text can be viewed in a reduced form way as capturing a wide range of slow moving underlying mechanisms which increase the supply of workers experienced in the new technology as the transition unfolds. Furthermore, it captures that many such mechanisms are increasing the share of existing technology workers.

D.3 Greater complementarities

TO BE COMPLETED.

D.4 Alternative bargaining protocol

TO BE COMPLETED.

E Social Planner Solution

Preliminaries. This section extends and generalizes Fukui and Mukoyama (2025) to the current setting with heterogeneous human capital, learning and technology choice. Let Λ_t denote the social value of a match, Θ_t the social value of a solo entrepreneur, and Ω_t the social value of an unemployed. Let the operator $\mathcal{A}_{ee}$ denote all within match state changes that do not involve any worker flows and captured in Equation (5). Likewise, define the operator $\mathcal{A}_{vv}$ and $\mathcal{A}_{uu}$.

Derivations. Using standard methods we can apply integration by parts to derive the Lagrangian of the social planner problem as

$$
\begin{aligned}
\mathcal{L} = \max_{v,\mathbb{1}^{EE},\mathbb{1}^{UE},\mathbb{1}^{A},\mathbb{1}^{X}} & \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \pi(z,h \mid \tau) dG_t^e(z,h \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \left(\pi(z,0 \mid \tau) - c(v_t(z \mid \tau))\right) dG_t^v(z \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int bh dG_t^u(h \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \mathcal{A}_{ee}\left[\Lambda_t\right](z,h \mid \tau) dG_t^e(z,h \mid \tau) dt \\
& + \int e^{-\rho t} \int \chi \mathbb{1}^{A}(z,h \mid \tau,\tau;t) \left(\mathbb{E}\left[\Lambda_t(z',h' \mid 1) \mid \tau,\tau\right] - \Lambda_t(z,h \mid 0)\right) dG_t^e(z,h \mid \tau) dt \\
& - \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \varkappa \mathbb{1}^{X}(z,h \mid \tau;t) \left(\Theta_t(z \mid \tau) + \Omega_t(h \mid \tau) - \Lambda_t(z,h \mid \tau)\right) dG_t^e(z,h \mid \tau) dt \\
& - \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \kappa \left(\Lambda_t(z,h \mid \tau) - \Theta_t(z \mid \tau) - \Omega_t(h \mid \tau)\right) dG_t^e(z,h \mid \tau) dt \\
& - \int e^{-\rho t} \sum_{\tau,\tau'\in\mathcal{T}_t} \iint \xi p(\theta_t) \left(\mathbb{1}^{EE}(z,h,z' \mid \tau,\tau';t) \frac{v_t(z' \mid \tau')}{\mathbb{V}_t} dG_t^v(z' \mid \tau') \right) \left(\Lambda_t(z,h \mid \tau) - \Theta_t(z \mid \tau)\right) dG_t^e(z,h \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \left(-\rho\Lambda_t(z,h \mid \tau) + \dot{\Lambda}_t(z,h \mid \tau)\right) dG_t^e(z,h \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \mathcal{A}_{vv}\left[\Theta_t\right](z \mid \tau) dG_t^v(z \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau,\tau'\in\mathcal{T}_t} \iint \xi p(\theta_t) \left(\mathbb{1}^{EE}(z',h',z \mid \tau';\tau,t) \left(\Lambda_t^H(z,h' \mid \tau,\tau') - \Theta_t(z \mid \tau)\right) dG_t^e(z',h' \mid \tau')\right) \frac{v_t(z \mid \tau)}{\mathbb{V}_t} dG_t^v(z \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau,\tau'\in\mathcal{T}_t} \iint p(\theta_t) \left(\mathbb{1}^{UE}(h,z \mid \tau',\tau;t) \left(\Lambda_t^H(z,h' \mid \tau,\tau') - \Theta_t(z \mid \tau) - \Omega_t(h' \mid \tau')\right) dG_t^u(h' \mid \tau')\right) \frac{v_t(z \mid \tau)}{\mathbb{V}_t} dG_t^v(z \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \left(-\rho\Theta_t(z \mid \tau) + \dot{\Theta}_t(z)\right) dG_t^v(z \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \mathcal{A}_{uu}\left[\Omega_t\right](h \mid \tau) dG_t^u(h \mid \tau) dt \\
& + \int e^{-\rho t} \sum_{\tau\in\mathcal{T}_t} \int \left(-\rho\Omega_t(h \mid \tau) + \dot{\Omega}_t(h \mid \tau)\right) dG_t^u(h \mid \tau) dt \\
& + \int e^{-\rho t} \lambda_t \left(\theta_t \mathbb{X}_t - \mathbb{V}_t\right) dt \\
& + \int e^{-\rho t} \mu_t \left(\sum_{\tau\in\mathcal{T}_t} \int dG_t^u(h \mid \tau) + \xi \sum_{\tau\in\mathcal{T}_t} \int dG_t^e(z,h \mid \tau) - \mathbb{X}_t \right) dt
\end{aligned}
$$

$$+\int e^{-\rho t}\omega_t\left(\sum_{\tau\in\mathcal{T}_t}\int v_t(z\mid\tau)dG^v_t(z\mid\tau)-\mathbb{V}_t\right)dt$$

$$+\int e^{-\rho t}\phi_t\left(\sum_{\tau\in\mathcal{T}_t}\int dG^e_t(z,h\mid\tau)+\sum_{\tau\in\mathcal{T}_t}\int dG^u_t(h\mid\tau)-1\right)dt$$

$$+\int e^{-\rho t}\varphi_t\left(\sum_{\tau\in\mathcal{T}_t}\int dG^e_t(z,h\mid\tau)+\sum_{\tau\in\mathcal{T}_t}\int dG^v_t(z\mid\tau)-M\right)dt,$$

subject to

$$\Lambda^H_t(z,h\mid\tau,\tau')=\mathbb{1}^A(z,h\mid\tau,\tau',t)\,\mathbb{E}\left[\Lambda_t(z',h'\mid 1)\mid z,h,\tau,\tau'\right]+\left(1-\mathbb{1}^A(z,h\mid\tau,\tau',t)\right)\Lambda_t(z,h\mid 0).$$

Social value of a match. By standard Calculus of Variation arguments the social value functions must satisfy the following set of PDEs

$$\begin{aligned}\rho\Lambda_t(z,h\mid\tau)-\partial_t\Lambda_t(z,h\mid\tau)=\ &\pi(z,h\mid\tau)+\mathcal{A}_{ee}\left[\Lambda_t\right](z,h\mid\tau)-\kappa\left(\Lambda_t(z,h\mid\tau)-\Theta_t(z\mid\tau)-\Omega_t(h\mid\tau)\right)\\&+\xi p(\theta_t)\sum_{\tau'\in\mathcal{T}_t}\int\mathbb{1}^{EE}\left(z,h,z'\mid\tau,\tau';t\right)\Delta^{EE}_t(z,h,z'\mid\tau,\tau')\frac{v_t(z'\mid\tau')}{\mathbb{V}_t}dG^v_t(z'\mid\tau')\\&+\xi\mu_t+\phi_t+\varphi_t\end{aligned}\tag{E.23}$$

where

$$\Delta^{EE}_t(z,h,z'\mid\tau,\tau')=\left(\Lambda^H_t(z',h\mid\tau',\tau)-\Theta_t(z'\mid\tau')\right)-\left(\Lambda_t(z,h\mid\tau)-\Theta_t(z\mid\tau)\right)$$

and Λ^H_t defined as above, and subject to the boundary conditions for adoption and endogenous separation subject to the boundary conditions for adoption and endogenous separation

$$\Lambda_t(z,h|\tau)\geq\Lambda^A_t(z,h\mid\tau,\tau),\qquad\text{and}\qquad\Lambda_t(z,h|\tau)\geq\Theta_t(z|\tau)+\Omega_t(h|\tau),$$

where the value of adopting is given by $\Lambda^A_t(z,h\mid\tau_f,\tau_w)\equiv\mathbb{E}\left[\Lambda_t(\tilde{z},\tilde{h}\mid 1)\mid z,h,\tau_f,\tau_w\right]$, where $\tilde{z}=\delta^{1-\tau_f}z$ and $\tilde{h}=\delta^{1-\tau_w}h$ and δ follows the stochastic process defined in (4). Equation (E.23) differs from the competitive equilibrium in (5) only in terms of the last two lines.

Social value of solo entrepreneur. The value of the solo entrepreneur is

$$\begin{aligned}\rho\Theta_t(z\mid\tau)-\partial_t\Theta_t(z,\mid\tau)&=\pi(z,0\mid\tau)-c(v_t(z\mid\tau))+\mathcal{A}_{vv}\left[\Theta_t\right](z\mid\tau)\\&+\xi p(\theta_t)\frac{v_t(z\mid\tau)}{\mathbb{V}_t}\sum_{\tau'\in\mathcal{T}_t}\iint\mathbb{1}^{EE}(z',h',z\mid\tau',\tau,t)\Delta_t^{EE}(z',h',z\mid\tau',\tau)dG_t^e(z',h'\mid\tau')\\&+p(\theta_t)\frac{v_t(z\mid\tau)}{\mathbb{V}_t}\sum_{\tau'\in\mathcal{T}_t}\int\mathbb{1}^{UE}(h',z\mid\tau',\tau,t)\Delta_t^{UE}(h',z\mid\tau',\tau)dG_t^u(h'\mid\tau')\\&+\omega_t v_t(z\mid\tau)+\varphi_t\end{aligned}\tag{E.24}$$

where

$$\Delta_t^{UE}(h,z'\mid\tau,\tau')=\Lambda_t^H(z',h\mid\tau',\tau)-\Theta_t(z'\mid\tau')-\Omega_t(h\mid\tau).$$

Equation (E.24) differs from the competitive equilibrium in (6) only in terms of the final line.

Value of unemployment. The value of unemployment given by

$$\begin{aligned}\rho\Omega_t(h\mid\tau)-\partial_t\Omega_t(h\mid\tau)=bh+\mathcal{A}_{uu}\left[\Omega_t\right](h\mid\tau)&\\+p(\theta_t)\sum_{\tau'\in\mathcal{T}_t}\int\frac{v_t(z'\mid\tau')}{\mathbb{V}_t}\mathbb{1}^{UE}\left(h,z'\mid\tau,\tau't\right)\Delta_t^{UE}(h,z'\mid\tau,\tau')g_t^v(z'\mid\tau')dz'&\\+\mu_t+\phi_t&\end{aligned}\tag{E.25}$$

This PDE in (E.25) for the social value of unemployment again differs from the competitive equilibrium counterpart by their final two lines.

Optimal policies. Optimal vacancy posting satisfies

$$\begin{aligned}c'(v_t(z\mid\tau))-\omega_t&=\xi p(\theta_t)\frac{1}{\mathbb{V}_t}\sum_{\tau'\in\mathcal{T}_t}\iint\mathbb{1}^{EE}\left(z',h',z\mid\tau',\tau,t\right)\Delta_t^{EE}\left(z',h',z\mid\tau',\tau,\right)dG_t^e(z',h'\mid\tau')\\&+p(\theta_t)\frac{1}{\mathbb{V}_t}\sum_{\tau'\in\mathcal{T}_t}\int\mathbb{1}^{UE}\left(h',z\mid\tau',\tau,t\right)\Delta_t^{UE}\left(h',z\mid\tau',\tau,\right)dG_t^u(h'\mid\tau').\end{aligned}$$

Optimal hiring and adoption policy then simply is given by

$$\mathbb{1}^{EE}(z,h,z' \mid \tau,\tau';t) = \begin{cases} 1 & \text{if } \Delta^{EE}(z,h,z' \mid \tau,\tau') \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{1}^{UE}(h,z' \mid \tau,\tau';t) = \begin{cases} 1 & \text{if } \Delta^{UE}(h,z' \mid \tau,\tau') \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{1}^{A}(z,h' \mid \tau,\tau';t) = \begin{cases} 1 & \text{if } \Lambda^{A}_t(z,h \mid \tau,\tau') \geq \Lambda_t(z,h' \mid 0) \\ 0 & \text{otherwise.} \end{cases}$$

$$\mathbb{1}^{X}(z,h \mid \tau;t) = \begin{cases} 1 & \text{if } \Lambda_t(z,h \mid \tau) \leq \Theta_t(z \mid \tau) + \Omega_t(h \mid \tau) \\ 0 & \text{otherwise.} \end{cases}$$

Congestion Lagrange multipliers satisfy

$$\begin{aligned} -\omega_t \mathbb{V}_t = {} & \xi p(\theta)\left(1 - \frac{p'(\theta_t)\theta_t}{p(\theta_t)}\right) \sum_{\tau,\tau' \in \mathcal{T}_t} \int \frac{v_t(z \mid \tau)}{\mathbb{V}_t} \Delta^{EE}_t\left(z',h',z \mid \tau',\tau\right)^+ dG^e_t(z',h' \mid \tau') dG^v_t(z \mid \tau) \\ & + p(\theta)\left(1 - \frac{p'(\theta_t)\theta_t}{p(\theta_t)}\right) \sum_{\tau,\tau' \in \mathcal{T}_t} \int \frac{v_t(z \mid \tau)}{\mathbb{V}_t} \Delta^{UE}_t\left(h',z \mid \tau',\tau\right)^+ dG^u_t(h' \mid \tau') dG^v_t(z \mid \tau) \end{aligned}$$

The worker search congestion is given by

$$\begin{aligned} -\mu_t \mathbb{X}_t = {} & \xi p(\theta_t)\left(\frac{p'(\theta_t)\theta_t}{p(\theta_t)}\right) \sum_{\tau,\tau' \in \mathcal{T}_t} \int \frac{v_t(z \mid \tau)}{\mathbb{V}_t} \Delta^{EE}_t(z',h',z \mid \tau',\tau)^+ dG^e_t(z',h' \mid \tau') dG^v_t(z \mid \tau) \\ & + p(\theta_t)\left(\frac{p'(\theta_t)\theta_t}{p(\theta_t)}\right) \sum_{\tau,\tau' \in \mathcal{T}_t} \int \frac{v_t(z \mid \tau)}{\mathbb{V}_t} \Delta^{UE}_t(h',z \mid \tau',\tau)^+ dG^u_t(h' \mid \tau') dG^v_t(z \mid \tau) \end{aligned}$$

Note that these congestion effects relate to elasticity of the matching function. This mirrors Hosios (1990).

Optimal taxes. These can be backed out by comparing the social planner and competitive equilibrium value functions.

F Computational Appendix

TO BE COMPLETED.

G Analytical Model

This section illustrates the trade-off between production complementarities, dynamic learning and match duration in an analytical model while capturing the main elements of the quantitative model. For most of the section we focus on the stylized case where there are two entrepreneurs and a single worker. This setting allows us to think through the key forces governing the optimal allocation of the worker between entrepreneurs as well as the optimal match duration that a social planner chooses. Towards the end of the section, we introduce a third entrepreneur C to understand the externalities and inefficiencies in the competitive equilibrium with Bertrand competition and learning.

There are two entrepreneurs A and B and a single worker. Time is continuous. Productivity is denoted by $z_A(t)$ and $z_B(t)$. The worker's human capital is denoted by $h(t)$. Suppose that $z_B(0) < h(0) < z_A(0)$, and $h(0) > 1$. How should the worker be allocated between entrepreneur A and B?

Production. Production features complementarities between z and h parameterized by χ. Specifically, output for firm $i \in \{A, B\}$ is given by

$$y_i(t) = \begin{cases} z_i(t) + h(t) + \chi z_i(t) h(t) & \text{if matched with worker} \\ z_i(t) & \text{otherwise.} \end{cases}$$

Learning dynamics. Learning happens from the more to the less knowledgeable agent and is increasing in the difference of the knowledge levels. Specifically, learning takes the form for the entrepreneur

$$\dot{z}_i(t) = \begin{cases} \alpha_1 \left[h(t) - z_i(t)\right]^+ & \text{if matched with worker} \\ 0 & \text{otherwise.} \end{cases}$$

For the worker matched with entrepreneur i human capital evolves as

$$\dot{h}(t) = \alpha_1 \left[z_i(t) - h(t)\right]^+ .$$

If a match were to last forever, the learning dynamics imply that the present value of productivity and human capital is a simple weighted average of the initial knowledge and the knowledge of the more knowledgeable peer.

Specifically,

$$\int_0^{+\infty} e^{-rt} h(t) dt = \frac{r}{r+\alpha_1}\frac{h(0)}{r} + \frac{\alpha_1}{r+\alpha_1}\frac{z_A(0)}{r} \qquad (h \text{ learns from } A)$$

$$\int_0^{+\infty} e^{-rt} z_B(t) dt = \frac{r}{r+\alpha_1}\frac{z_B(0)}{r} + \frac{\alpha_1}{r+\alpha_1}\frac{h(0)}{r} \qquad (B \text{ learns from } h)$$

$$\int_0^{+\infty} e^{-rt} z_A(t) dt = \frac{z_A(0)}{r} \qquad (A \text{ does not learn})$$

Once and for all allocation. First, we consider the optimal time zero allocation of the worker assuming that it is permanent. The gain from allocating the worker to A comes from two sources: (i) a production misallocation effect modulated through the complementarity parameter χ as the production function is supermodular and (ii) a learning effect of h learning from the more productive firm. The gain from allocating the worker to B is a pure learning effect, i.e. from raising the productivity of the laggard. The following Lemma formulizes this.

Lemma 5. *The net present value of output from allocating to entrepreneur A and B, respectively, are*

$$V_A = \frac{z_A(0)}{r} + \frac{z_B(0)}{r} + \left(\frac{r}{r+\alpha_1}\frac{h(0)}{r} + \frac{\alpha_1}{r+\alpha_1}\frac{z_A(0)}{r}\right) + \chi\left(\frac{r}{r+\alpha_1}\frac{z_A(0)h(0)}{r} + \frac{\alpha_1}{r+\alpha_1}\frac{z_A(0)^2}{r}\right)$$

$$V_B = \frac{z_A(0)}{r} + \left(\frac{r}{r+\alpha_1}\frac{z_B(0)}{r} + \frac{\alpha_1}{r+\alpha_1}\frac{h(0)}{r}\right) + \frac{h(0)}{r} + \chi\left(\frac{r}{r+\alpha_1}\frac{z_B(0)h(0)}{r} + \frac{\alpha_1}{r+\alpha_1}\frac{h(0)^2}{r}\right).$$

It is optimal to allocate the worker to the less productivity entrepreneur B if and only if

$$\underbrace{\frac{\alpha_1}{r+\alpha_1}\left(\frac{h(0)}{r} - \frac{z_B(0)}{r}\right)}_{\text{gain } B \text{ learning}} - \underbrace{\frac{\alpha_1}{r+\alpha_1}\left(\frac{z_A(0)}{r} - \frac{h(0)}{r}\right)}_{\text{gain } h \text{ learning}} > \underbrace{\chi\frac{(z_A(0) - z_B(0))\,h(0)}{r+\alpha_1}}_{\text{static misallocation}} + \underbrace{\chi\frac{\alpha_1}{r+\alpha_1}\left(\frac{z_A(0)^2}{r} - \frac{h(0)^2}{r}\right)}_{\text{dynamic misallocation}} \tag{G.26}$$

The left hand side in G.26 motivates worker mobility and diffusion [better explanation of intuition needed]. If the learning gains for B from h exceed the learning gains of h from A, then there is a motive for allocating the worker to B. The right hand side denotes the sum of static and dynamic production misallocation costs. For $\chi > 0$ there are production complementarities between z and h. The first expression captures the losses from misallocating at initial knowledge levels. The second dynamic expression captures that these losses are compounded by the fact that the worker becomes more knowledgeable over time at A.

It is instructive to consider limiting cases. If there is no learning or complementarities in production are large, i.e. $\alpha_1/\chi \to 0$, then the condition can never be satisfied given $z_A(0) > \max\{z_B(0), h(0)\}$ as only the misallocation terms remain. When complementarities in production are shut off, i.e. $\chi \to 0$, then the above condition reduces to

$$\left(\frac{h(0)}{r}-\frac{z_B(0)}{r}\right)-\left(\frac{z_A(0)}{r}-\frac{h(0)}{r}\right)>0.$$

Intuitively, with no consideration for production complementarities the social planner wants to allocate the worker to maximize total knowledge and therefore exploit all potential learning gains. Away from this limit case the social planner trades off production value and learning gains.

Optimal match duration. The previous equation motivates the following thought exercise. What is the optimal duration T that the worker should stay at firm A before moving to B? The social planner maximizes

$$\max_T \int_0^T e^{-rt}\left(z_A(0)+h(t)+z_B(0)\right)dt+\chi\int_0^T e^{-rt}z_A(0)h(t)dt+e^{-rt}V_B(h(T)),$$

where the continuation value of allocating the worker with human capital to firm B is given by

$$V_B(h)=\frac{z_A(0)}{r}+\frac{h}{r}+\left(\frac{r}{r+\alpha_1}\frac{z_B(0)}{r}+\frac{\alpha_1}{r+\alpha_1}\frac{h}{r}\right)+\chi h\left(\frac{r}{r+\alpha_1}\frac{z_B(0)}{r}+\frac{\alpha_1}{r+\alpha_1}\frac{h}{r}\right).$$

Its derivative is

$$V_B'(h)=\frac{1}{r}+\frac{\alpha_1}{r+\alpha_1}\frac{1}{r}+\chi\frac{z_B(0)}{r+\alpha_1}+2\chi\frac{\alpha_1}{r+\alpha_1}\frac{h}{r}.$$

The first order condition with respect to T is given by

$$\underbrace{z_A(0)+h(T)+z_B(0)+\chi z_A(0)h(T)+}_{\text{(1) flow return of delay}}\quad\underbrace{V_B'(h(T))h'(T)}_{\text{(2) increased learning gains for } B}\quad-\underbrace{rV_B(h(T))}_{\text{(3) cost of delay}}=0.$$

Noting that $h'(T)=\alpha_1\left(z_A(0)-h(T)\right)$. The above boils down to finding the optmal threshold level of h^* that satisfies the above equation. Substituting in for the functional forms we obtain the quadratic equation

$$\begin{aligned}0=-\frac{\alpha_1}{r+\alpha_1}\left(h^*-z_B(0)\right)+\chi\frac{r}{r+\alpha_1}\left(z_A(0)-z_B(0)\right)h^*&\\+\chi\frac{\alpha_1}{r+\alpha_1}\left(z_A(0)-h^*\right)h^*&\\+\alpha_1\left(\frac{1}{r}+\frac{\alpha_1}{r+\alpha_1}\frac{1}{r}+\chi\frac{z_B(0)}{r+\alpha_1}+2\chi\frac{\alpha_1}{r+\alpha_1}\frac{h^*}{r}\right)\left(z_A(0)-h^*\right).&\end{aligned}$$

A sufficient condition for finite T to exist is $z_A(0)<\alpha_1/(\chi r)$. Proof: Plugin in for $h=z_A(0)$. Likewise, we can plug in for $h=z_B(0)$ and show that the $LHS>0$. Then, can use the intermediate value theorem.

Once we have found the optimal h^* we can back out the optimal match duration

$$T(h^*) = \frac{1}{\alpha_1} \log\left(\frac{z_A(0) - h(0)}{z_A(0) - h^*}\right).$$

Discussion. Let us interpret these motives. Delaying the reallocation comes with two benefits. First, the worker is allocated to the more productive entrepreneur A. In the presence of production complementarities this yields large contemporaneous aggregate output. Second, insofar as that entrepreneur A is more productive than the worker there are additional gains from the worker remaining at A as this allows for the human capital to increase. Crucially, this gives another motive for match duration in the model. On the other hand, the cost of delay stems from the missed learning opportunities by entrepreneur B.

Competitive equilibrium. I now illustrate the sources of inefficient worker allocations in a setting of the bargaining protocol by Postel-Vinay and Robin (2002). Entrepreneurs Betrtrand compete on wages for the worker such that the worker moves to the entrepreneur where they have the greatest marginal surplus. As a result, the bargaining protocol is bilaterally efficient and all learning gains between A and B are fully internalized. Therefore, to be study the role of knowledge spillovers we introduce a third entrepreneur C. Anticipating the results below, the primary inefficiency comes from entrepreneurs A and B not internalizing how their hiring decisions affect the endogenous creation of knowledge which may spill over to a prospective poachee C.

For illustrative purposes, let $z_B(0) \leq z_C(0) < h(0) < z_A(0)$. B and C have the potential to learn from h and h has the potential to learn from A. To isolate the key inefficiency we assume the following deterministic labor market setting. At time 0 the worker is allocated to entrepreneur B and there is an opportunity of A to poach the worker. After a time interval T there is a meeting between the worker and the entrepreneur C who gets the opportunity to poach the worker. The following event tree summarizes the event tree.

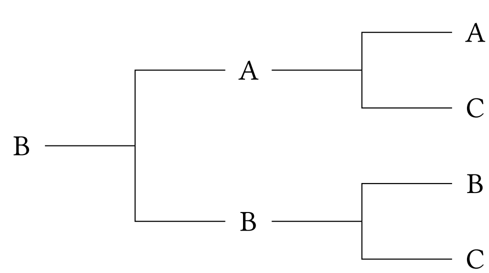


Figure G.9: Event sequences in labor market.

Before turning to the inefficiencies, it is instructive to first describe the allocations in the competitive equilibrium the sequential auction bargaining protocol by Postel-Vinay and Robin (2002). If the worker is at A inititially and after a time period T moves to C then based on the bargaining protocol the entrepreneur and worker value are, respectively,

$$J_0(A \mid h) = \int_0^T e^{-rt} \left(z_A(0) + h(t) + \chi z_A(0)h(t) - w_t\right) + \int_T^{+\infty} e^{-rt} z_A(0) dt$$
$$W_0(h \mid A) = \int_0^T e^{-rt} w_t dt + W_T(h \mid C)$$

where the sequential auction protocol implies that the worker extracts the value of the past employer A from the new employer C. That is, the maximum value A would be willing to pay to employ the worker $W_T(h \mid C) = \int_T^{+\infty} e^{-rt} \left(z_A(0) + h(t) + \chi z_A(0)h(t) - z_A(0)\right) dt$. By the Bertrand assumption this is the wage that C offers if poaching was optimal. Combined, the total private surplus is given by the following expression that is independent of whether C poaches the worker

$$S_0(A, h) = J_0(A \mid h) + W_0(h \mid A) = \int_0^{+\infty} e^{-rt} \left(z_A(0) + h(t) + \chi z_A(0)h(t) - z_B(0)\right) dt.$$

Likewise, the total private surplus of the worker remaining at B is

$$S_0(B, h) = \int_0^{+\infty} e^{-rt} \left(z_B(t) + h(0) + \chi z_B(t)h(0) - z_B(0)\right) dt.$$

Again by Bertrand competition entrepreneurs are able to pay up to their full surplus to employ the worker. As utility is perfectly transferable the worker will therefore move to the entrepreneur with the greatest surplus. It follows that in the competitive equilibrium the worker remains at entrepreneur B if and only if

$$\int_0^{+\infty} e^{-rt} \left(z_B(t) + h(0) + \chi z_B(t)h(0) - z_B(0)\right) dt > \int_0^{+\infty} e^{-rt} \left(z_A(0) + h(t) + \chi z_A(0)h(t) - z_B(0)\right) dt.$$

We note that this is identical to the condition discussed in the bilateral case and all results carry over. It follows that in absence of a third entrepreneur C the competitive equilibrium is perfectly efficient.

Next let us turn to the social planner solution

Lemma 6. *The social value of moving the worker to A is given by*

$$W_A^{SP} = \underbrace{\int_0^{+\infty} e^{-rt}\left(z_A(0) + h(t) + \chi z_A(0)h(t)\right)dt + \int_0^{\infty} e^{-rt} z_B(0)dt + \int_0^{+\infty} e^{-rt} z_C(0)dt}_{\textit{private value}}$$

$$+ \underbrace{\int_T^{+\infty} e^{-rt}\left(z_C(t \mid A) + h(T \mid A) + \chi z_C(t \mid A)h(T \mid A) - z_C(0)\right)dt}_{\textit{added surplus from moving worker to C (including learning)}}$$

$$- \underbrace{\int_T^{+\infty} e^{-rt}\left(h(t) + \chi z_A(0)h(t)\right)dt}_{\textit{social cost of moving worker away from A}}$$

Some discussion here. E.g. that sum of last two lines is positive. That matters how much residual learning. Also on χ such that large χ accentuates the production misallocation.

Likewise,

$$W_B^{SP} = \underbrace{\int_0^{+\infty} e^{-rt}\left(z_A(0)dt + z_B(t \mid B) + h(0) + \chi z_B(t \mid B)h(0)\right)dt + \int_0^{+\infty} e^{-rt} z_C(0)dt}_{\textit{private value}}$$

$$+ \underbrace{\int_T^{+\infty} e^{-rt}\left(z_C(t \mid B) + h(0) + \chi z_C(t \mid B)h(0) - z_C(0)\right)dt}_{\textit{added surplus from moving worker to C (including learning)}}$$

$$- \underbrace{\int_T^{+\infty} e^{-rt}\left(z_B(t \mid B) + h(0) + \chi z_B(t \mid B)h(0) - z_B(T \mid B)\right)dt}_{\textit{social cost of moving worker away from B}}$$

It then follows that the wedge between the social planner and the competitive equilibrium are

$$
\begin{aligned}
e^{rT}\left(W_B^{SP}-W_A^{SP}-(W_B^{CE}-W_A^{CE})\right) = & \underbrace{-\int_T^{+\infty} e^{-rt}\left(z_C(t\mid A)-z_C(t)\right)dt}_{\textit{greater learning for } C \textit{ if first at } A} \\
& \underbrace{-\chi\int_T^{+\infty} e^{-rt}\left(z_C(t\mid A)h(T\mid A)-z_C(t\mid B)h(0)\right)dt}_{\textit{misallocation due to greater learning if first at } A} \\
& \underbrace{-\int_T^{+\infty} e^{-rt}\left(z_B(t\mid B)-z_B(T\mid B)\right)dt}_{\textit{foregone learning at } B \textit{ if move to } C} \\
& \underbrace{+\int_T^{+\infty} e^{-rt}\left(h(t\mid A)-h(T\mid A)\right)dt}_{\textit{foregone learning of } h \textit{ if move to } C} \\
& \underbrace{+\chi\int_T^{+\infty} e^{-rt}\left(z_A(0)h(t\mid A)-z_B(t\mid B)h(0)\right)dt}_{\textit{misallocation due to greater productivities at } A}
\end{aligned}
$$

Combining terms

$$
\begin{aligned}
e^{rT}\left(W_B^{SP}-W_A^{SP}-(W_B^{CE}-W_A^{CE})\right) = & \underbrace{-\frac{\alpha_1}{r+\alpha_1}\left[\left(\frac{h(T\mid A)}{r}-\frac{h(0)}{r}\right)-\left(\left(\frac{z_A(0)}{r}-\frac{h(T\mid A)}{r}\right)-\left(\frac{h(0)}{r}-\frac{z_B(T)}{r}\right)\right)\right]}_{\textit{(1)}} \\
& \underbrace{+\chi\frac{r}{r+\alpha_1}\frac{(z_A(0)-z_C(0))\,h(T\mid A)-(z_B(T)-z_C(0))\,h(0)}{r}}_{(2)} \\
& \underbrace{+\chi\frac{\alpha_1}{r+\alpha_1}\left(\frac{z_A(0)^2-h(T\mid A)^2}{r}\right)}_{(3)}
\end{aligned}
$$

where (1) denotes the net learning gains and (2) and (3) the net misallocation.

Comment. TO BE COMPLETED.